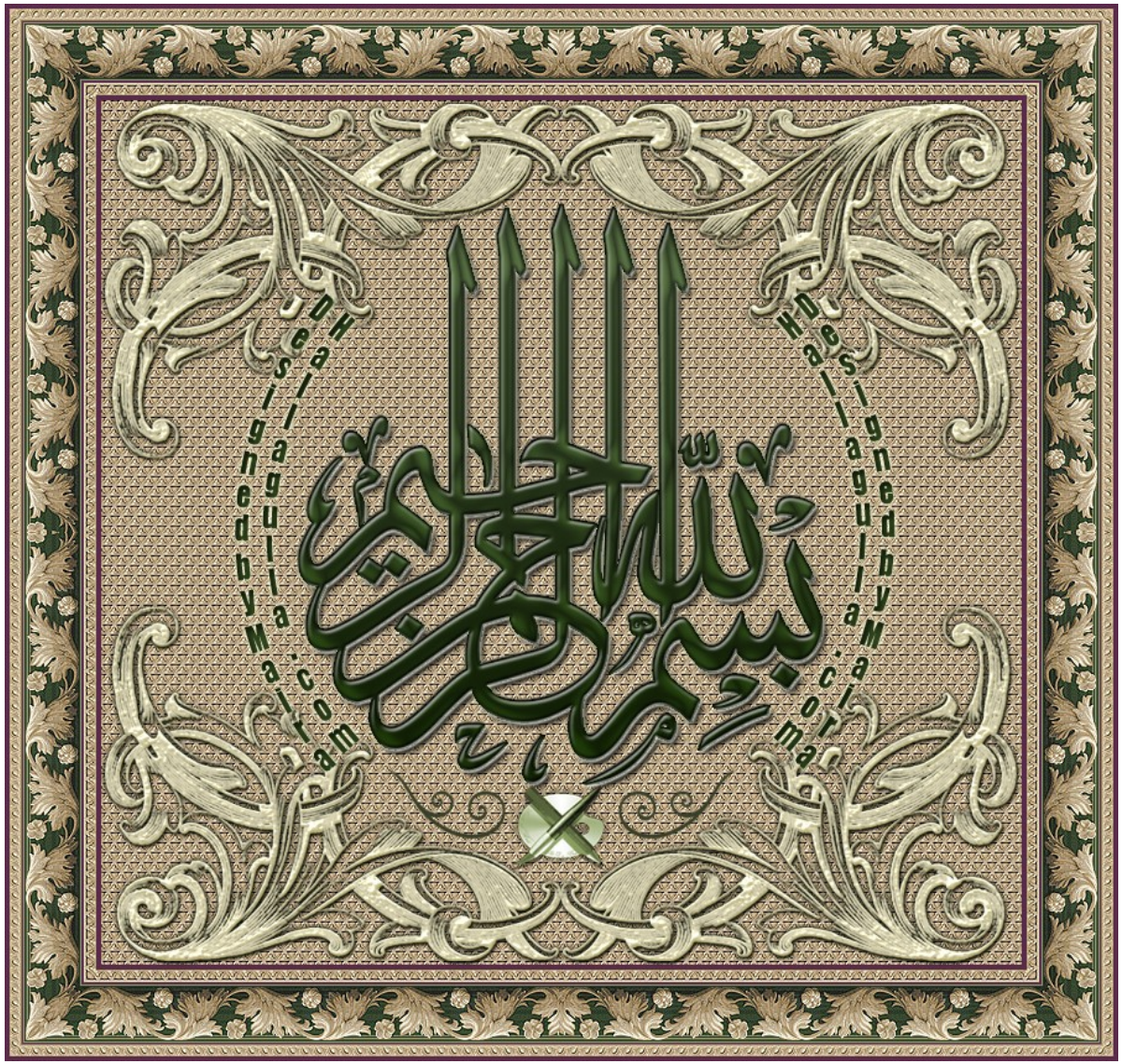
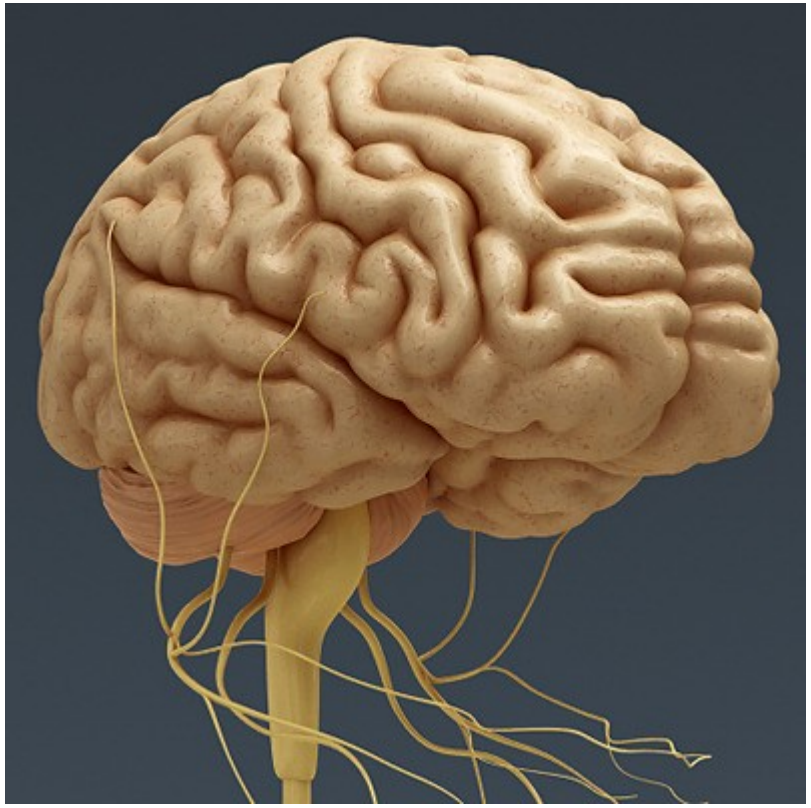
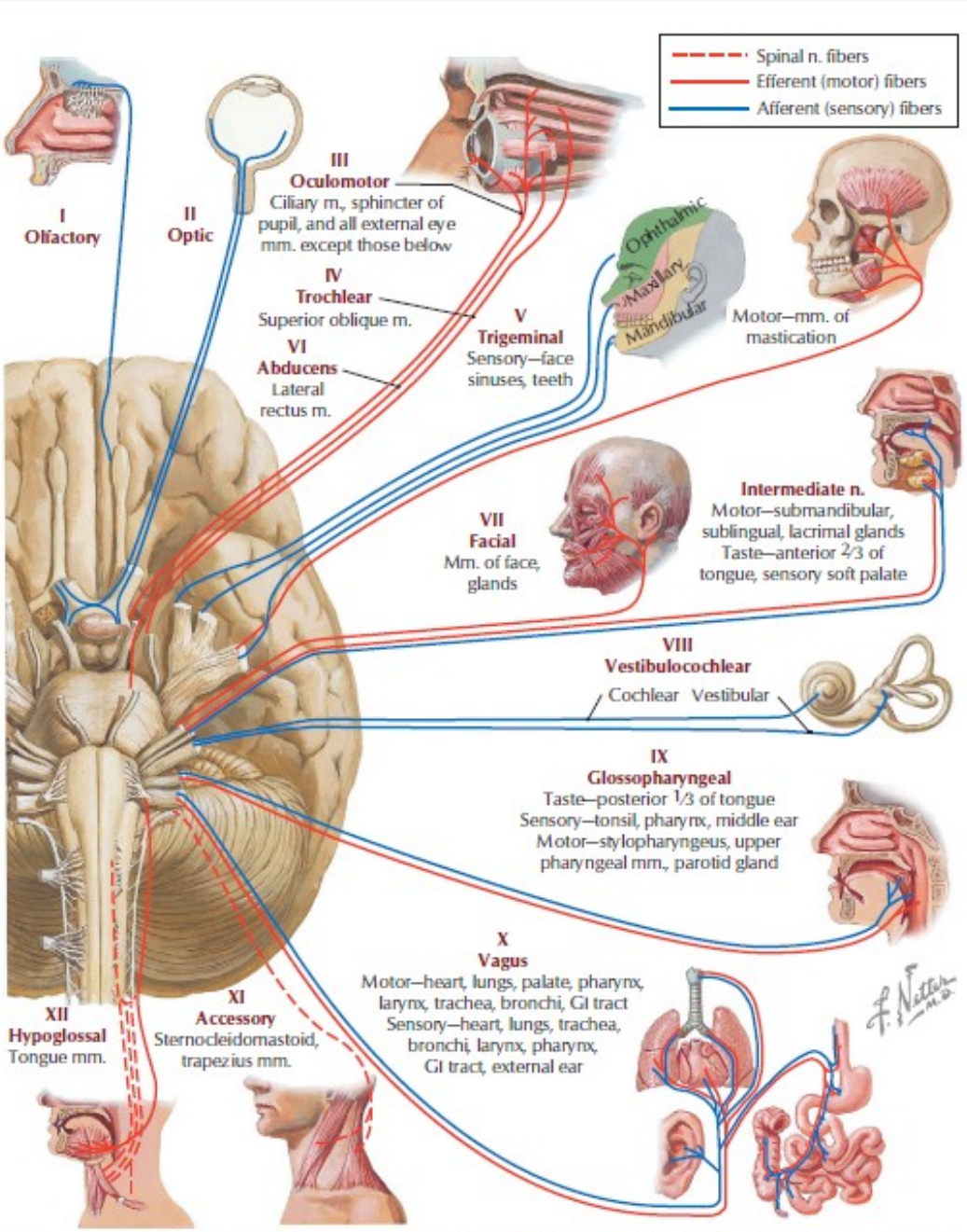

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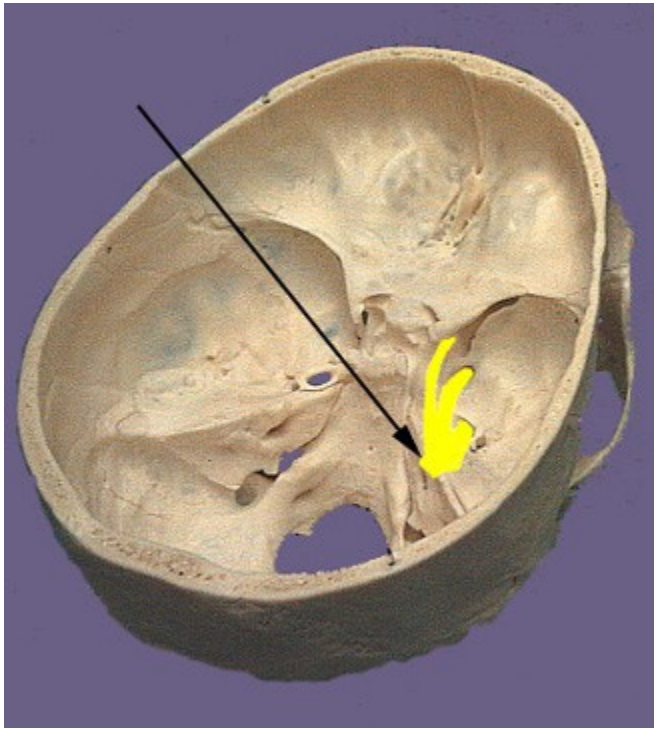


The Cranial Nerves

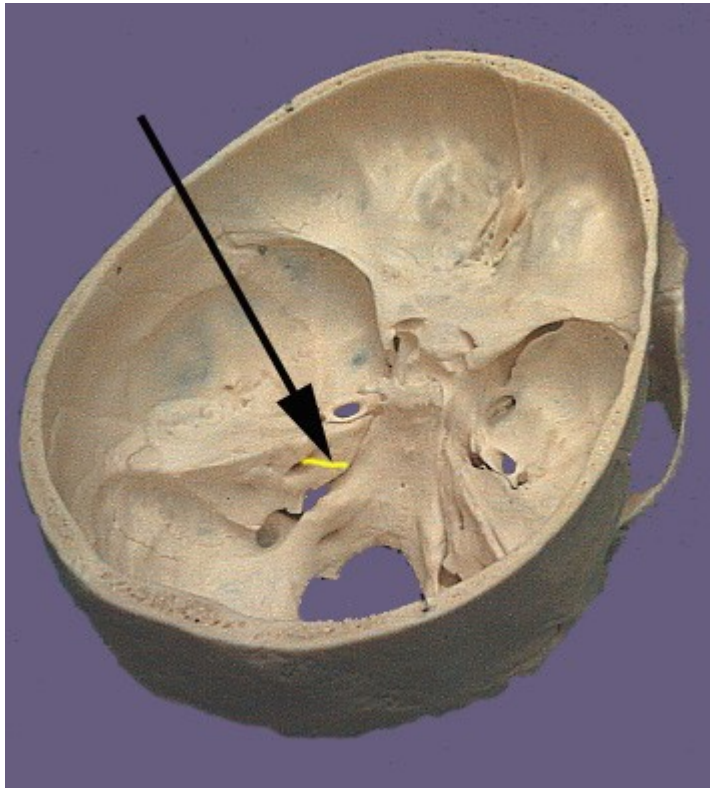
cranial nerves

- I Olfactory nerve(snuff)
- II Optic nerve(visible)
- III Oculomotor nerve
- IV Trochlear nerve(pulley)
- V Trigeminal nerve
- VI Abducent nerve
- VII Facial nerve
- VIII Vestibulocochlear nerv
- IX Glossopharyngeal nerve
- X Vagus nerve(wandering)
- XI Accessory nerve
- XII Hypoglossal nerve





CN v



CN vii

II. **BASIS OF CLASSIFICATION** - three letter system.

A. First letter

G = General = types of neurons found both in spinal nerves and cranial nerves.

S = Special = types of neurons only found in cranial nerves not spinal nerves.

B. Second letter

S = Somatic = types of neurons innervating structures derived from somites.

V = Visceral = types of neurons innervating gut, structures derived from or associated with gut and branchial arches; also vascular system, smooth muscle, internal organs and glands.

C. Third letter

A = Afferent = sensory neurons.

E = Efferent = motor neurons to skeletal and smooth muscle; also secretomotor neurons to glands.

Functional components

- **General somatic afferent fibers (GSA):** transmit exteroceptive and proprioceptive impulses from head and face to somatic sensory nuclei
- **Special somatic afferent fibers (SSA):** transmit sensory impulses from special sense organs of vision, equilibrium and hearing to the brain
- **General visceral afferent fibers (GVA):** transmit interoceptive impulses from the viscera to the visceral sensory nuclei
- **Special visceral afferent fibers (SVA):** transmit sensory impulses from special sense organs of smell and taste to the brain
- **General somatic efferent fibers (GSE):** innervate skeletal muscles of eye and tongue
- **Special visceral efferent fibers (SVE):** transmit motor impulses from the brain to skeletal muscles derived from brachial (gill) arches of embryo. These include the muscles of mastication, facial expression and swallowing
- **General visceral efferent fibers (GVE):** transmit motor impulses from the general visceral motor nuclei and relayed in parasympathetic ganglions. The postganglionic fibers supply cardiac muscles， smooth muscles and glands

Classification of cranial nerves

- **Sensory cranial nerves**: contain only afferent (sensory) fibers
  - **IOlfactory nerve**
  - **IIOptic nerve**
  - **VIII Vestibulocochlear nerve**
- **Motor cranial nerves**: contain only efferent (motor) fibers
  - **III Oculomotor nerve**
  - **IV Trochlear nerve**
  - **VIAbducent nerve**
  - **XI Accessory nerv**
  - **XII Hypoglossal nerve**
- **Mixed nerves**: contain both sensory and motor fibers---
  - **VTrigeminal nerve,**
  - **VII Facial nerve,**
  - **IXGlossopharyngeal nerve**
  - **XVagus nerve**

Trigeminal nerve

Components of fibers

- **SVE fibers**: originate from motor nucleus of trigeminal nerve, and supply masticatory muscles
- **GSA fibers**: transmit facial sensation to sensory nuclei of trigeminal nerve, the GSA fibers have their cell bodies in trigeminal ganglion, which lies on the apex of petrous part of temporal bone

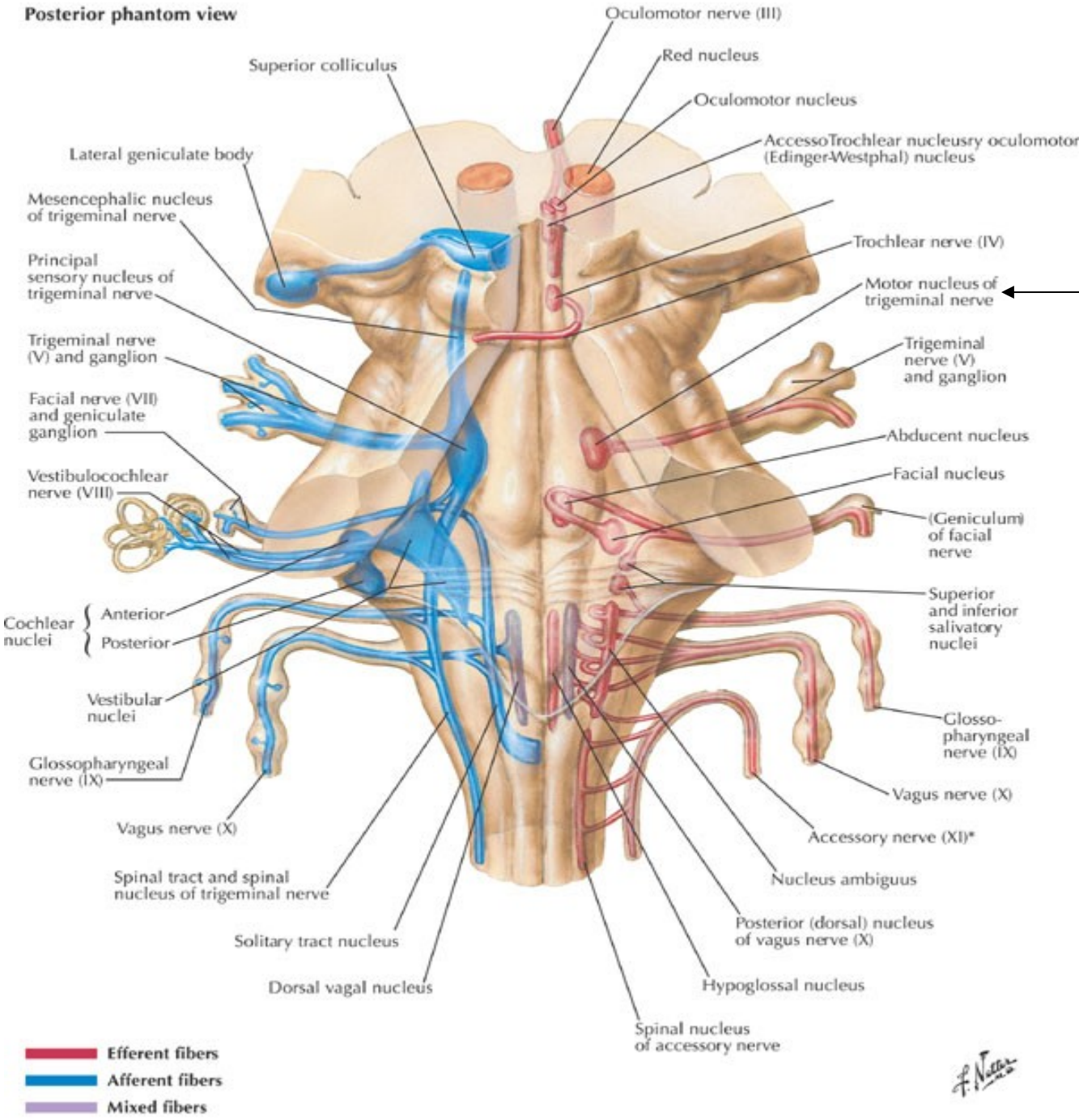
It has four nuclei

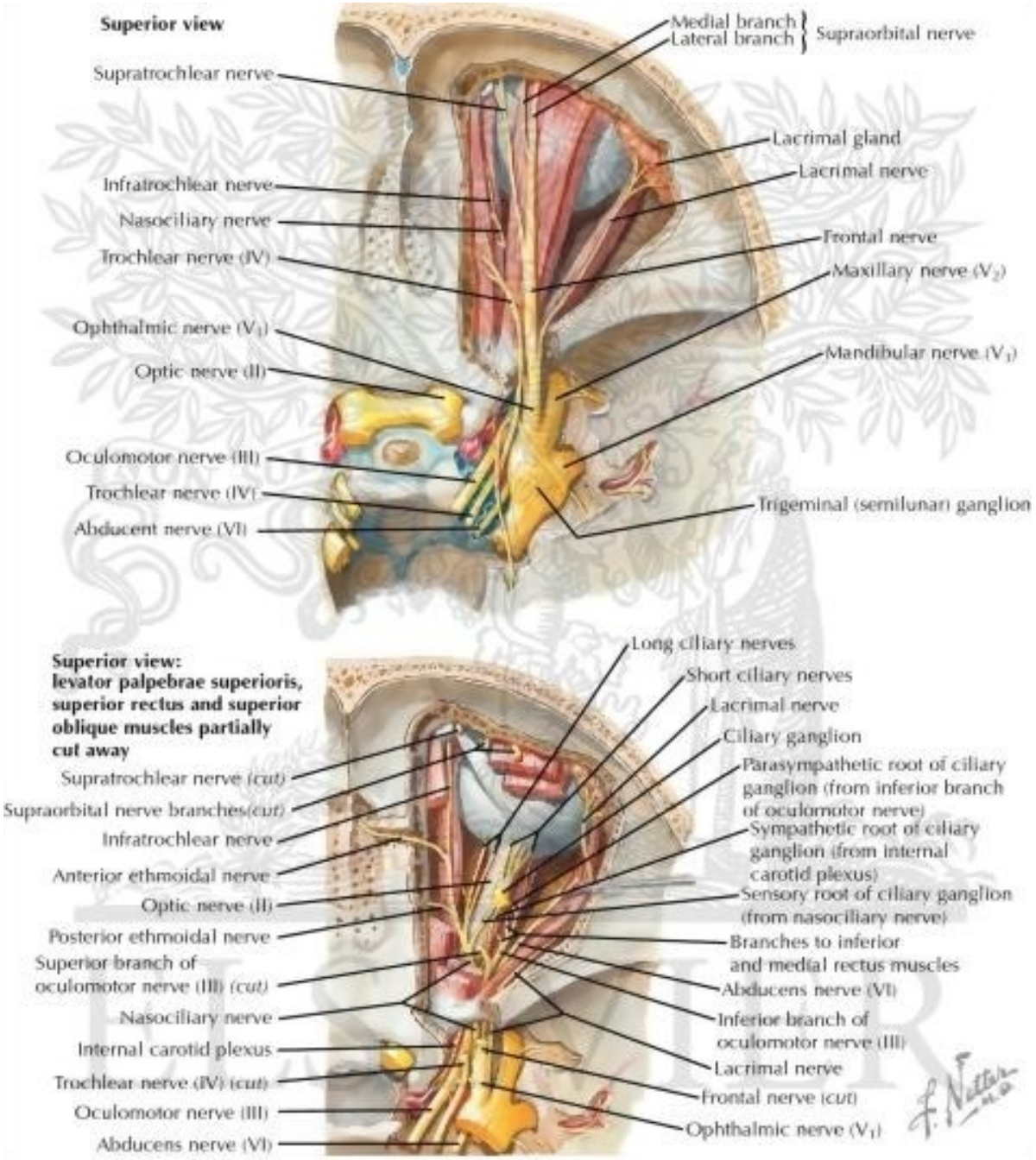
Main sensory nucleus lies posterior to pons & is continuos below with spinal nucleus

Spinal nucleus extends inferiorly through the whole length of Medulla oblongata & into upper part of spinal cord till c2 segment

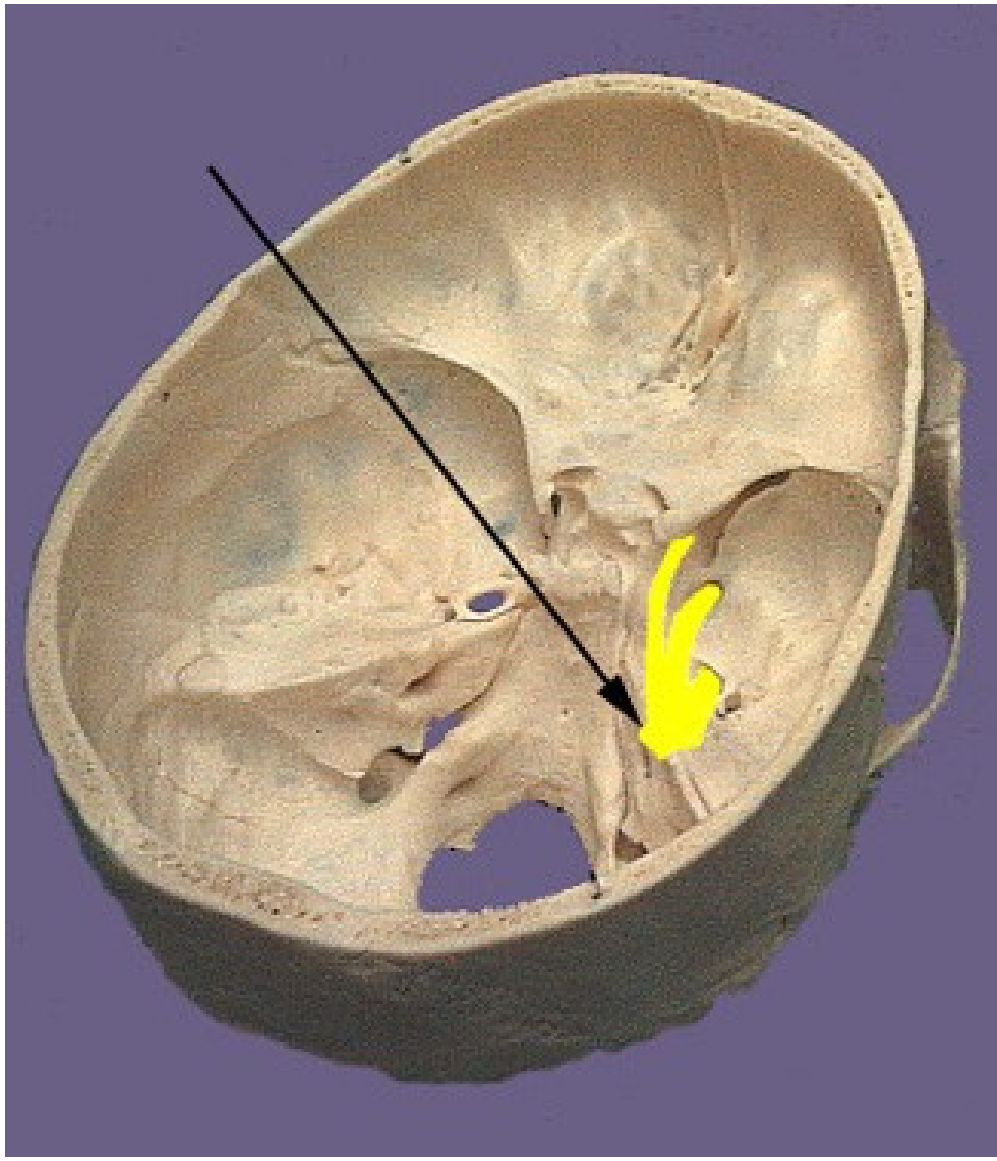
Mesencephalic nucleus situated in lateral part of cerebral aquaduct & extends inferiorly into the pons as far as the main sensory nucleus

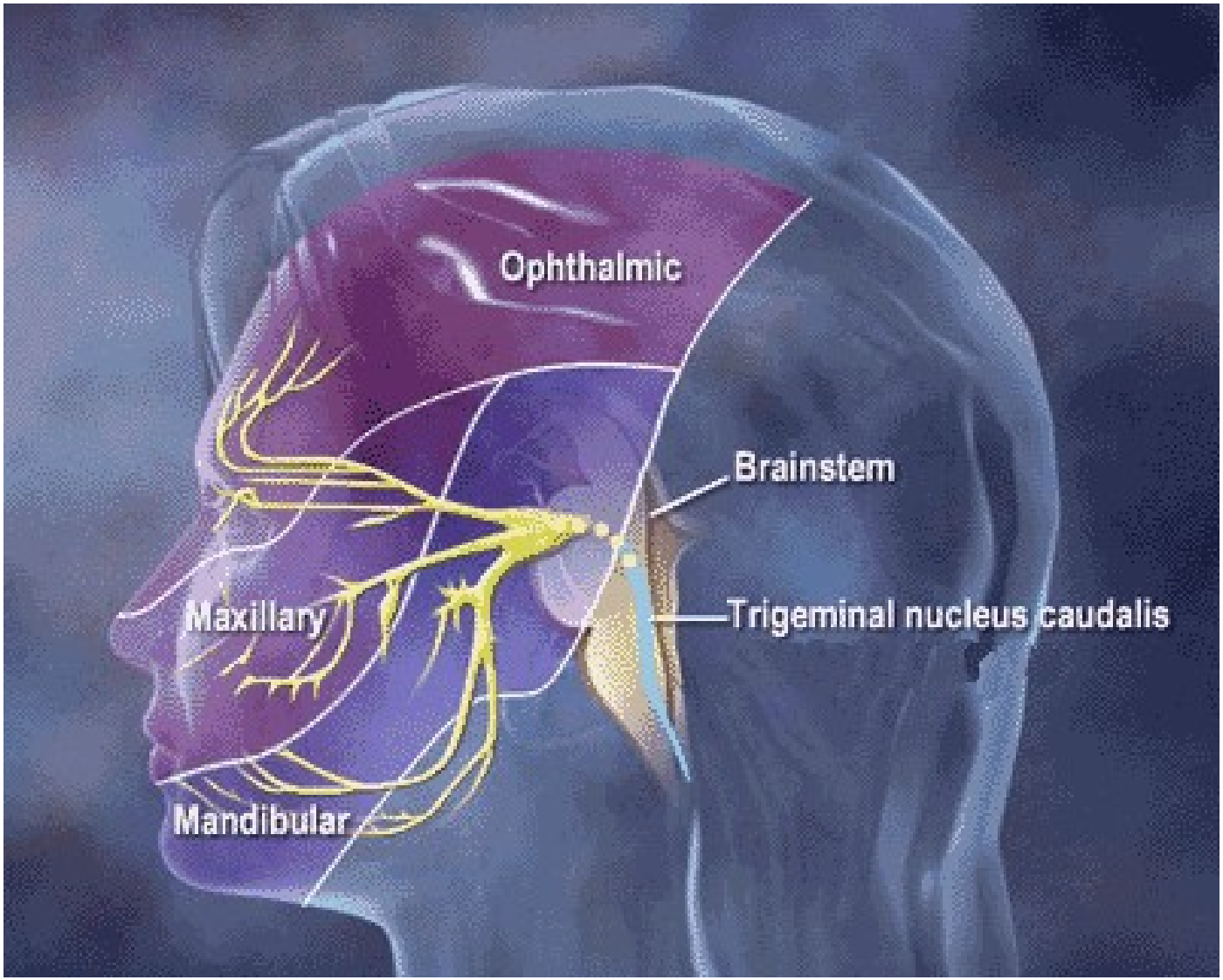
Motor nucleus is situated in the pons medial to main sensory nucleus.

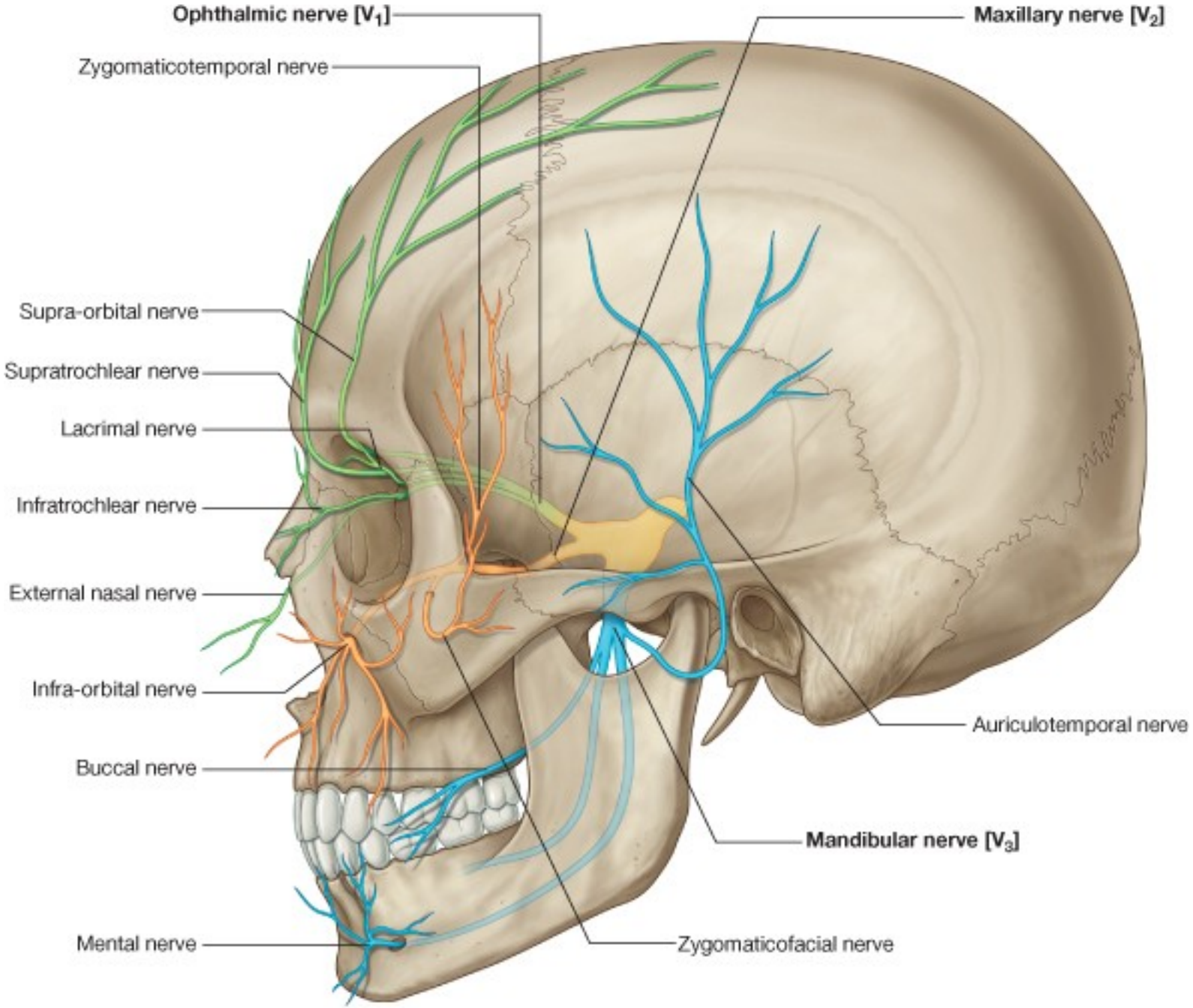



Superior view
Medial branch
Lateral branch
Supraorbital nerve
Supratrochlear nerve
Lacrimal gland
Lacrimal nerve
Infratrochlear nerve
Nasociliary nerve
Frontal nerve
Trochlear nerve (IV)
Maxillary nerve (V2)
Ophthalmic nerve (V1)
Optic nerve (II)
Mandibular nerve (V3)
Oculomotor nerve (III)
Trochlear nerve (IV)
Trigeminal (semilunar) ganglion
Abducent nerve (VI)
Superior view:
levator palpebrae superioris,
superior rectus and superior
oblique muscles partially
cut away
Long ciliary nerves
Short ciliary nerves
Lacrimal nerve
Ciliary ganglion
Supratrochlear nerve (cut)
Parasympathetic root of ciliary
ganglion (from inferior branch
of oculomotor nerve)
Supraorbital nerve branches (cut)
Infratrochlear nerve
Sympathetic root of ciliary
ganglion (from internal
carotid plexus)
Anterior ethmoidal nerve
Sensory root of ciliary ganglion
(from nasociliary nerve)
Optic nerve (II)
Posterior ethmoidal nerve
Branches to inferior
and medial rectus muscles
Superior branch of
oculomotor nerve (III) (cut)
Abducens nerve (VI)
Nasociliary nerve
Inferior branch of
oculomotor nerve (III)
Internal carotid plexus
Lacrimal nerve
Trochlear nerve (IV) (cut)
Frontal nerve (cut)
Oculomotor nerve (III)
Ophthalmic nerve (V1)
Abducens nerve (VI)

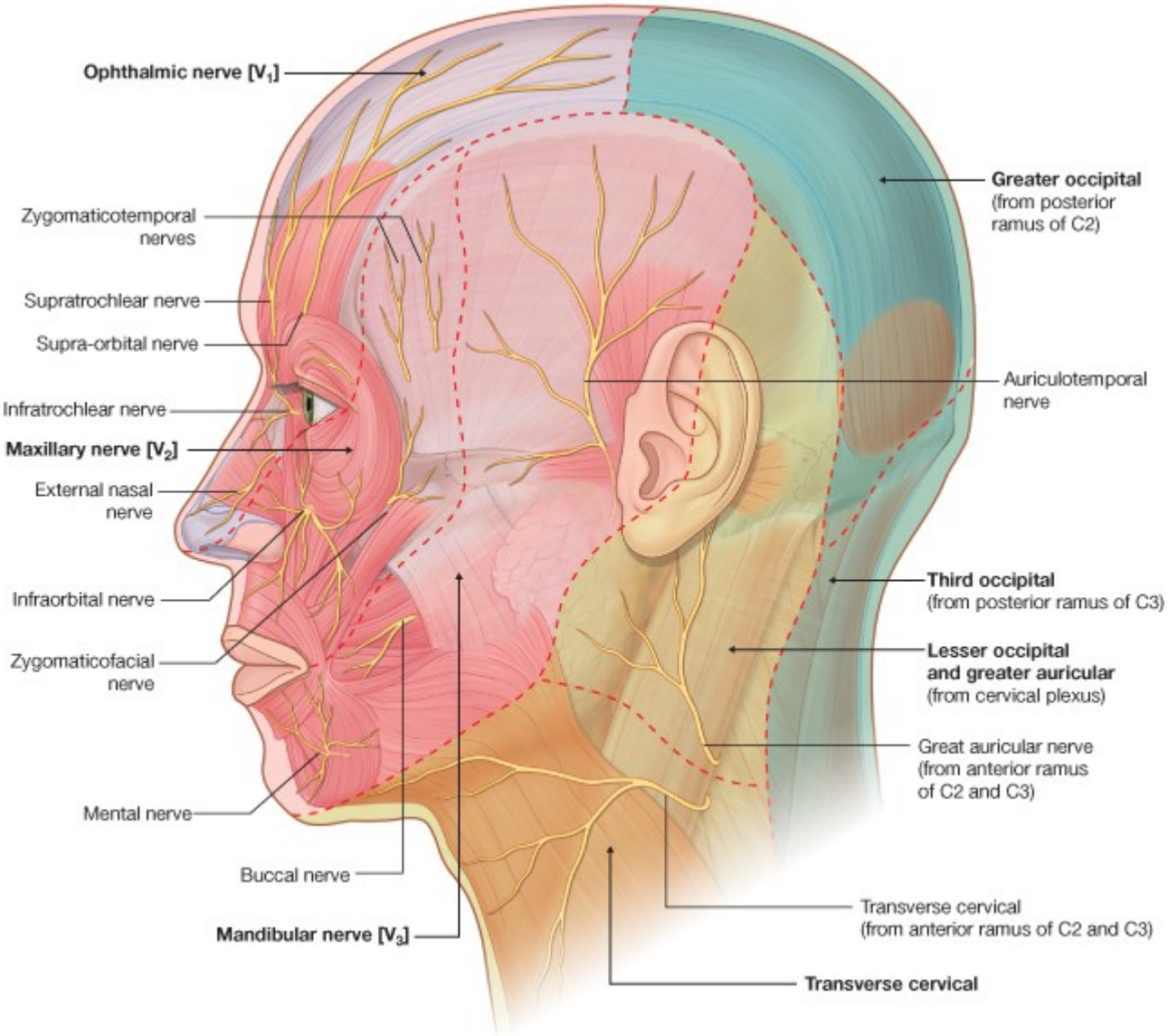
The **trigeminal nerve** is the largest cranial nerve and is the great sensory nerve of the head and face, and the motor nerve of the muscles of mastication. At the level of the midpons, trigeminal nerve emerges, by a small *motor* and a large *sensory root*—the former being situated in front of and medial to the latter .**trigeminal nerve** leaves the anterior aspect of pons as a small motor large sensory root . Nerve passes forward out of posterior cranial fossa & rests on upper surface of apex of petrous part of temporal bone in middle cranial fossa. Sensory root expands (trigerminal ganglion) **Ophthalmic nerve** , **Maxillary nerve** & **Mandibular nerve** arise from anterior border of ganglion..




Ophthalmic
Brainstem
Maxillary
Trigeminal nucleus caudalis
Mandibular



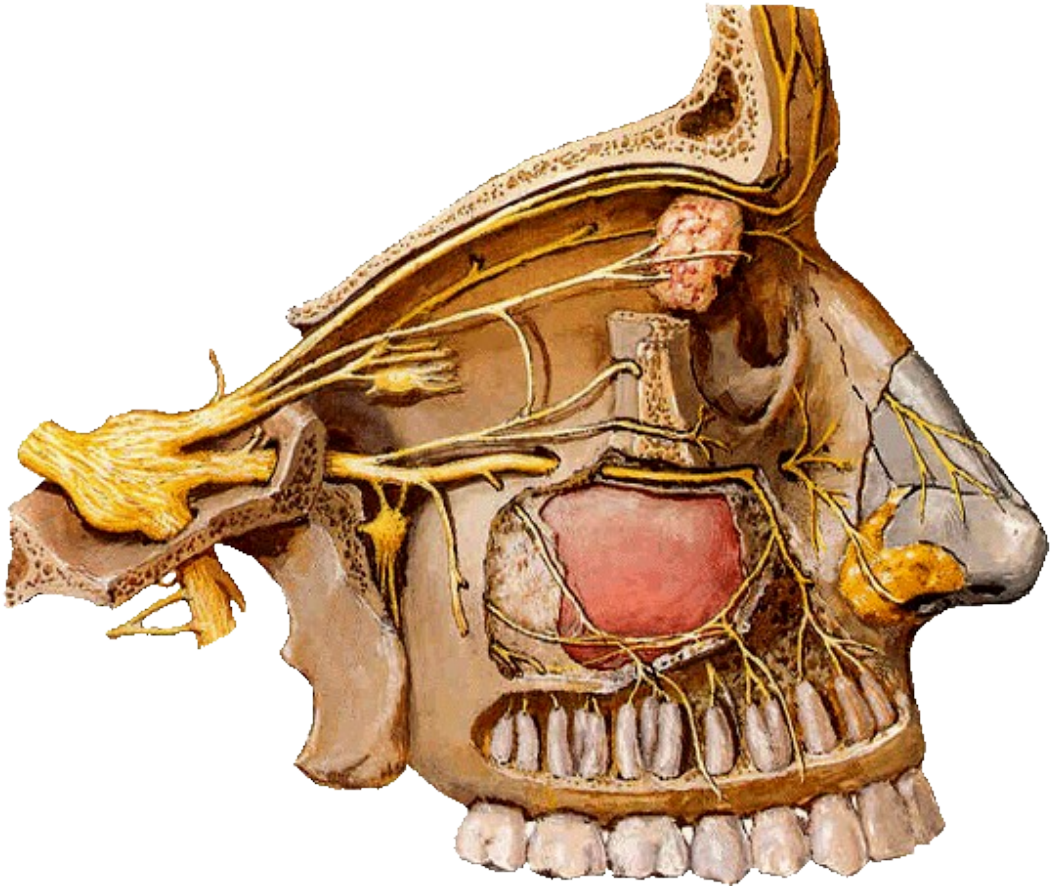
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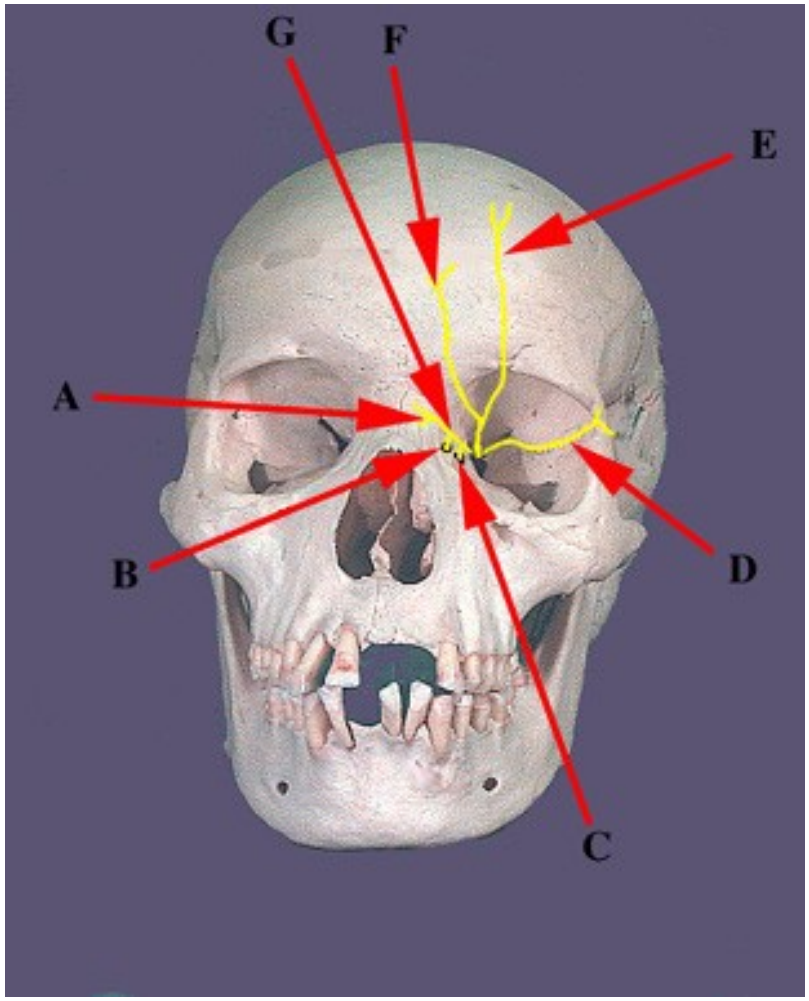


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Branches

- **Ophthalmic nerve** (V1, sensory) leave the skull through the superior orbital fissure, to enter orbital cavity
- Branches
  - Frontal nerve
    - Supratrochlear nerve
      Supraorbital nerve
      Lacrimal nerve
      Nasociliary nerve





A. Infratrochlear
B. Anterior Ethmoid
C. Posterior Ethmoid
D. Lacrimal
E. Supraorbital
F. Supratrochlear
G. Nasociliary

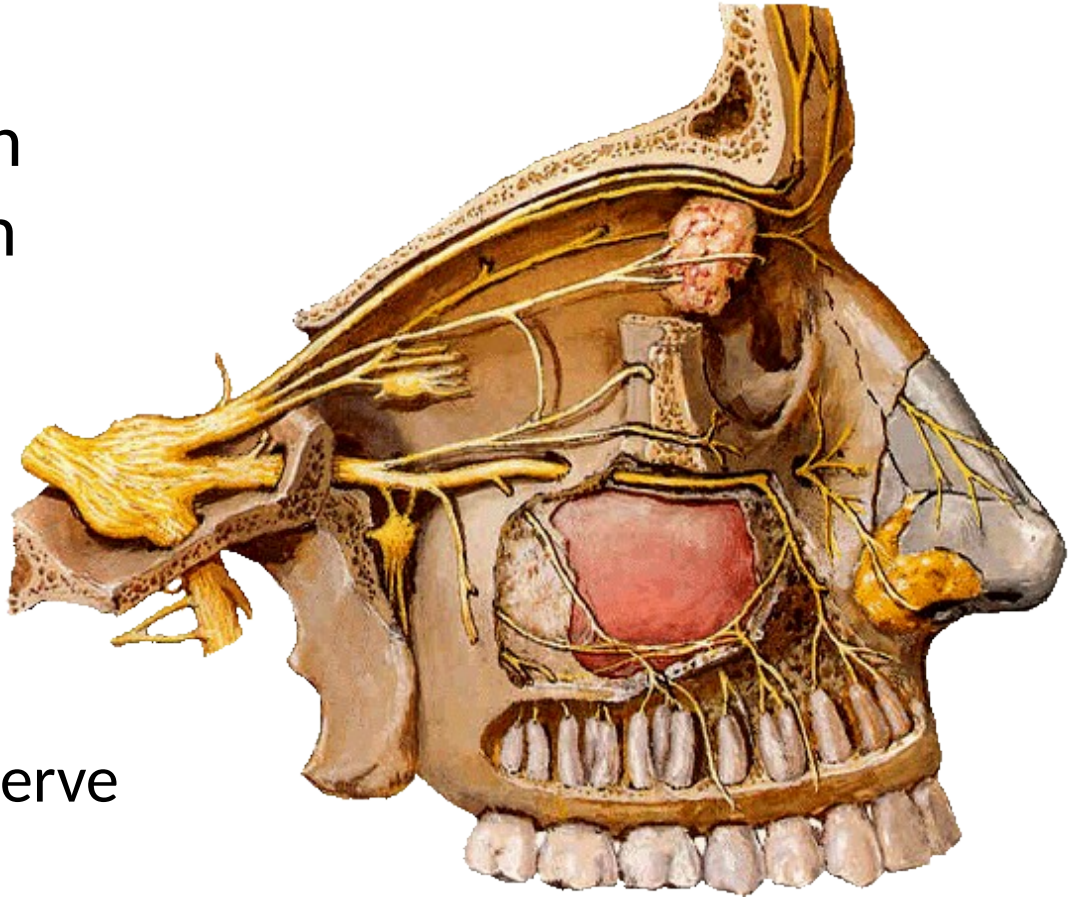
Distribution:

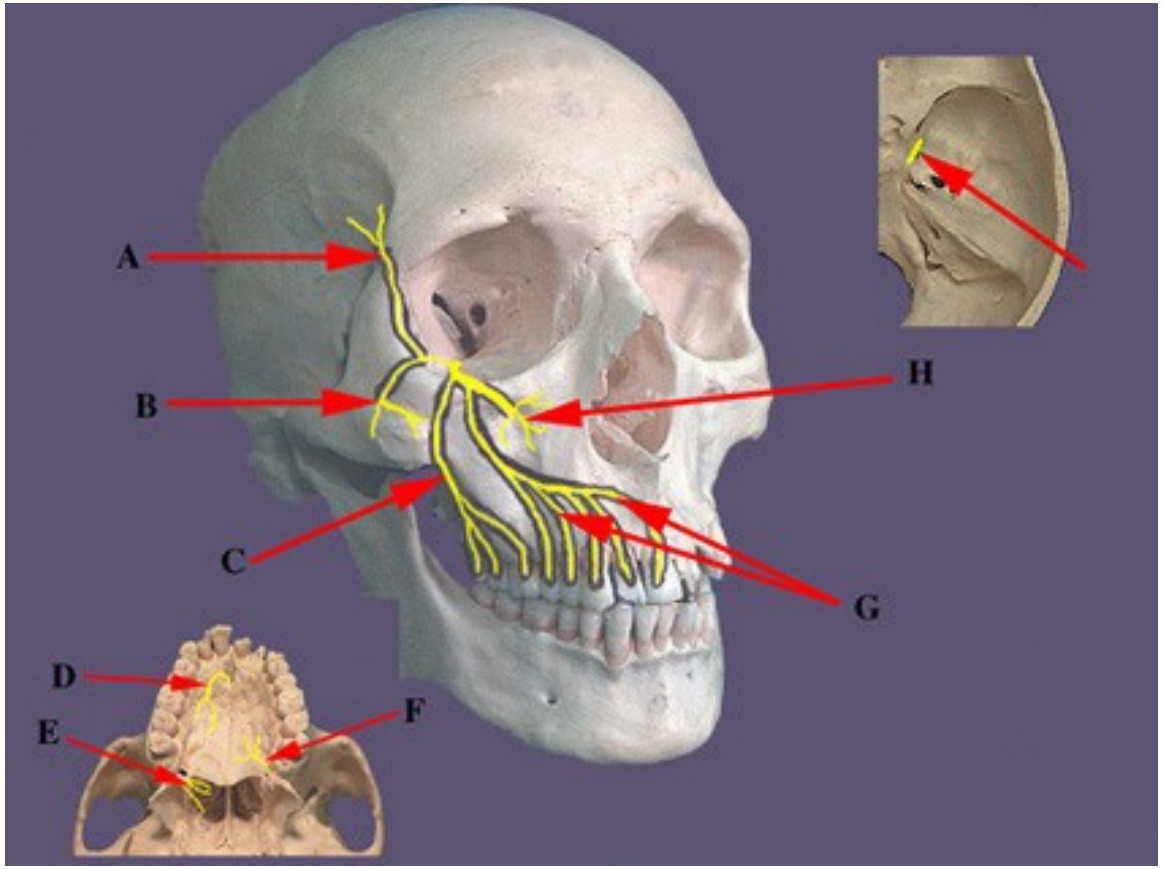
- Sensation from cerebral dura mater
- Visual organ
- Mucosa of nose
- Skin above the eye and back of nose



Maxillary nerve (V2, sensory)

- Leave skull through foramen rotundum
- Branches
  - Infraorbital nerve
  - Zygomatic nerve Superior alveolar nerve
  - Pterygopalatine nerve





A. Zygoticaticotemporal
B. Zygomaticofacial
C. Post. Sup. Alveolar Brs
D. Nasopalatine
E. Greater Palatine
F. Lesser Palatine
G. Mid. & Ant. Alveolar Brs
H. Infraorbital

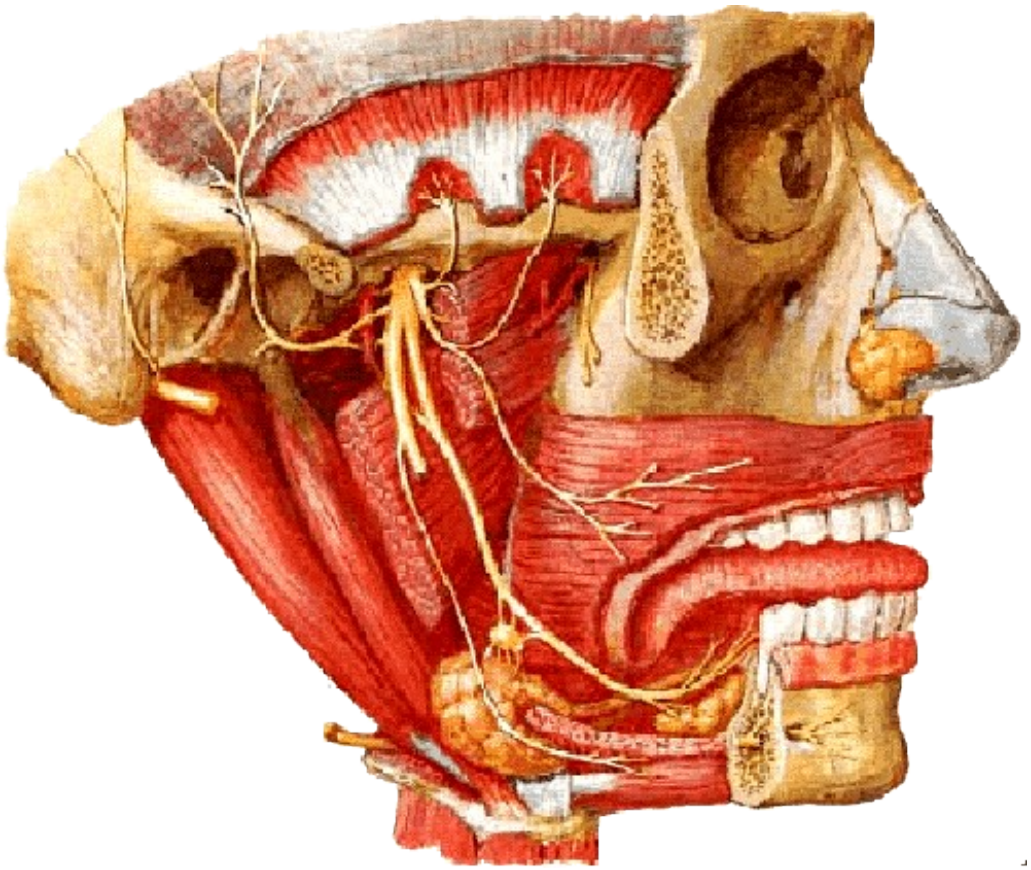
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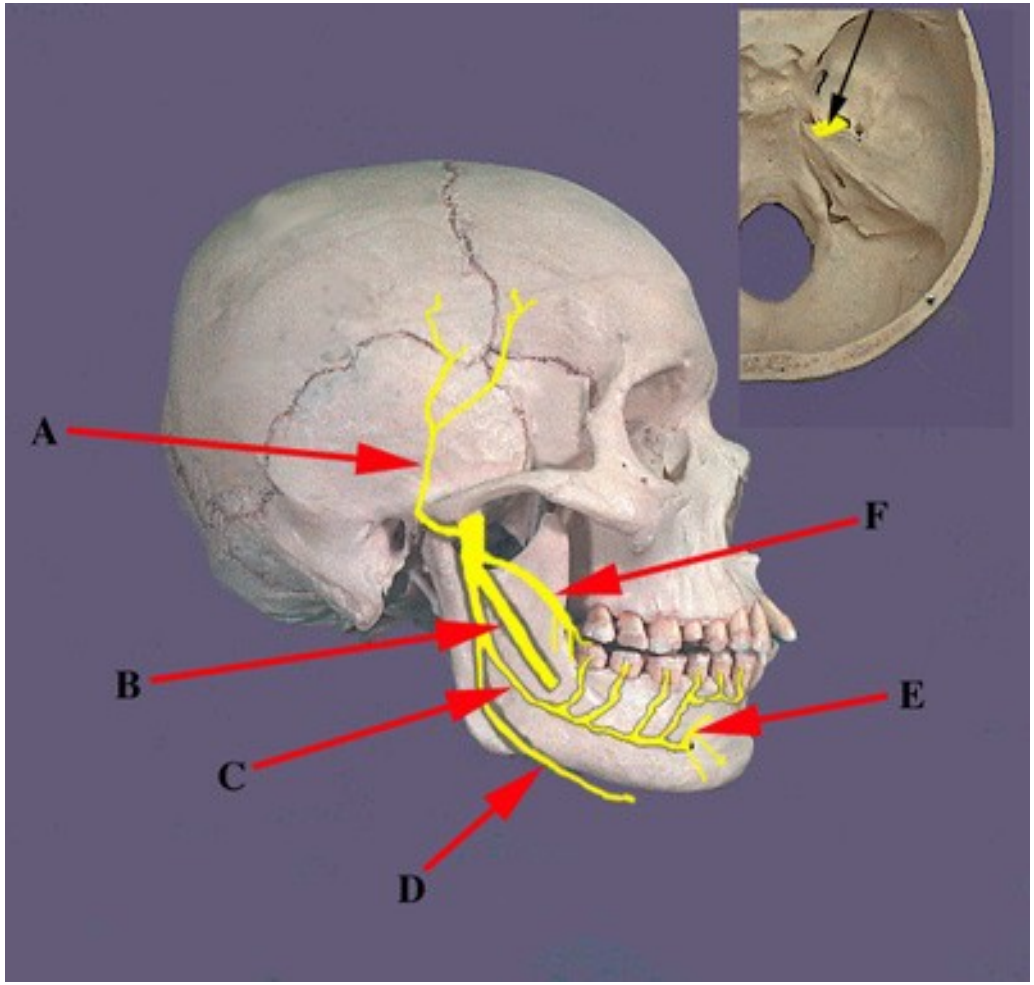
- Sensation from cerebral dura mater
- Maxillary teeth
- Mucosa of nose and mouth
- Skin between eye and mouth



Mandibular nerve (V3, mixed) Leave the skull through the **foramen ovale** to enter the infratemporal fossa

- Branches
  - **Auriculotemporal nerve**
  - **Buccal nerve**
  - **Lingual nerve**
  - **Inferior alveolar nerve**
  - **Nerve of masticatory muscles**





A. Auriculotemporal
B. Lingual
C. Inferior Alveolar
D. N. to the Mylohyoid
E. Mental
F. Buccal

Distribution:

- Sensation from cerebral dura mater
- Teeth and gum of lower jaw
- Mucosa of floor of mouth
- Anterior 2/3 of tongue
- Skin of auricular and temporal regions and below the mouth
- Motor to mastication muscles, mylohyoid, and anterior belly of digastric



Facial nerve (VII)

Components of fibers

- **SVE** fibers originate from nucleus of facial nerve, and supply facial muscles
- **GVE** fibers derived from superior salivatory nucleus and relayed in pterygopalatine ganglion and submandibular ganglion. The postganglionic fibers supply lacrimal, submandibular and sublingual glands
- **SVA** fiber from taste buds of anterior two-thirds of tongue which cell bodies are in the geniculate ganglion of the facial nerve and end by synapsing with cells of nucleus of solitary tract
- **GSA** fibers from skin of external ear

Components of fibers

- **SVE Special visceral efferent fibers** originate from motor nucleus of facial nerve (the **facial nucleus)**, and supply **branchiomotor** innervation facial muscles
- The lower half of the face is controlled by the contralateral cerebral hemisphere. The upper half is bilaterally controlled and therefore not paralyzed by an “upper motor neuron” lesion.

- The nervus intermedius consists of the axons of the **GVE General visceral efferent fibers** derived from superior salivatory nucleus and relayed in pterygopalatine ganglion and submandibular ganglion. The postganglionic fibers supply as **parasympathetic** (secretomotor) innervation to the lacrimal, submandibular and sublingual glands

Components of fibers

- **SVA Special visceral afferent fibers** from taste buds of anterior two-thirds of tongue which cell bodies are in the **geniculate ganglion** the only sensory ganglion of the facial nerve and end by synapsing with cells of nucleus of solitary tract

- GSA sensation from the area posterior to the ear, whereas others carry GVA sensation from the nasal cavity and soft palate.

Components of fibers

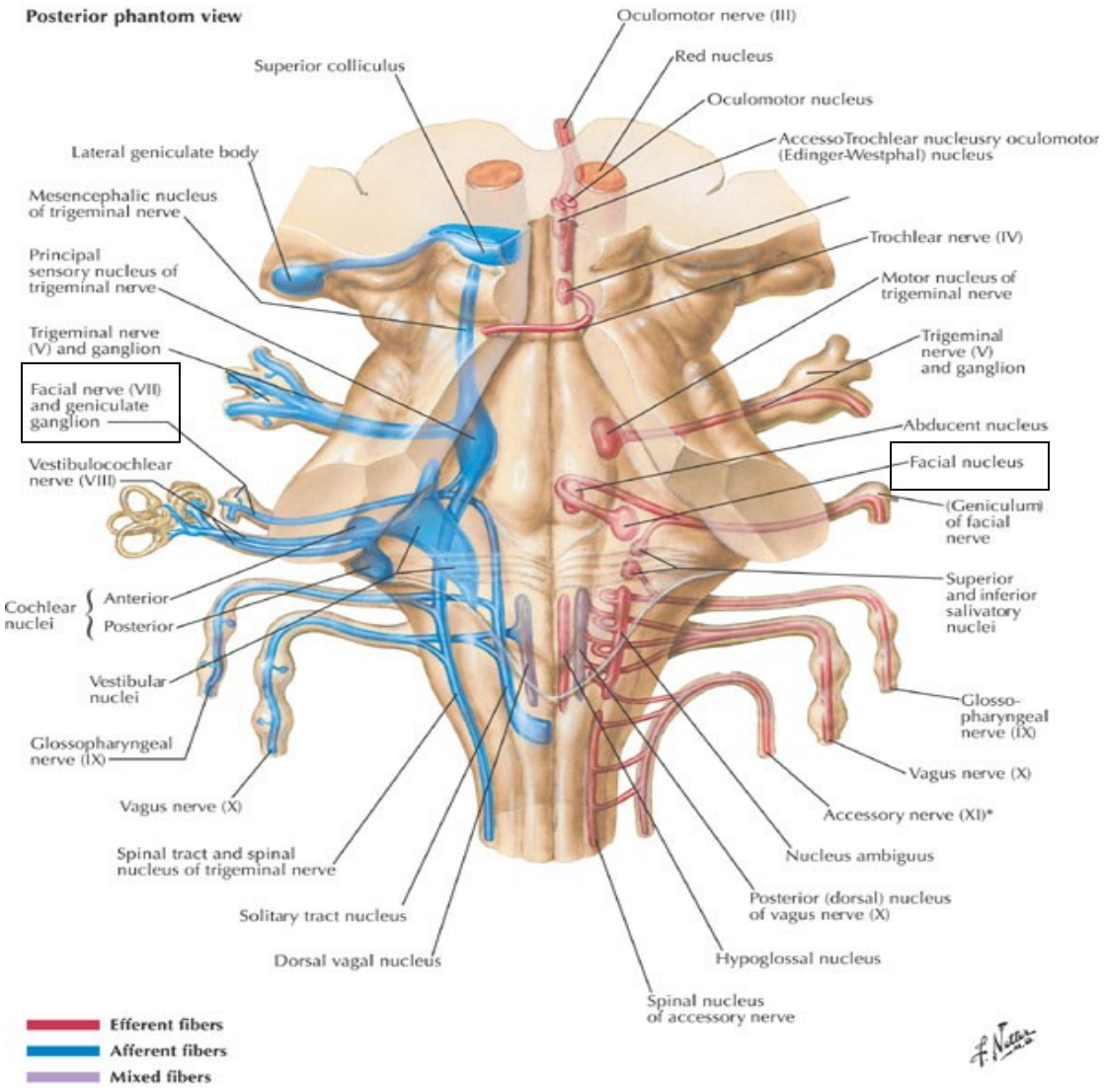
- **GSA General somatic afferent fibers**
- All general somatic sensory fibers from cranial nerve ganglia (V and IX; some from VII and X) end in trigeminal nuclei.

- PARTS

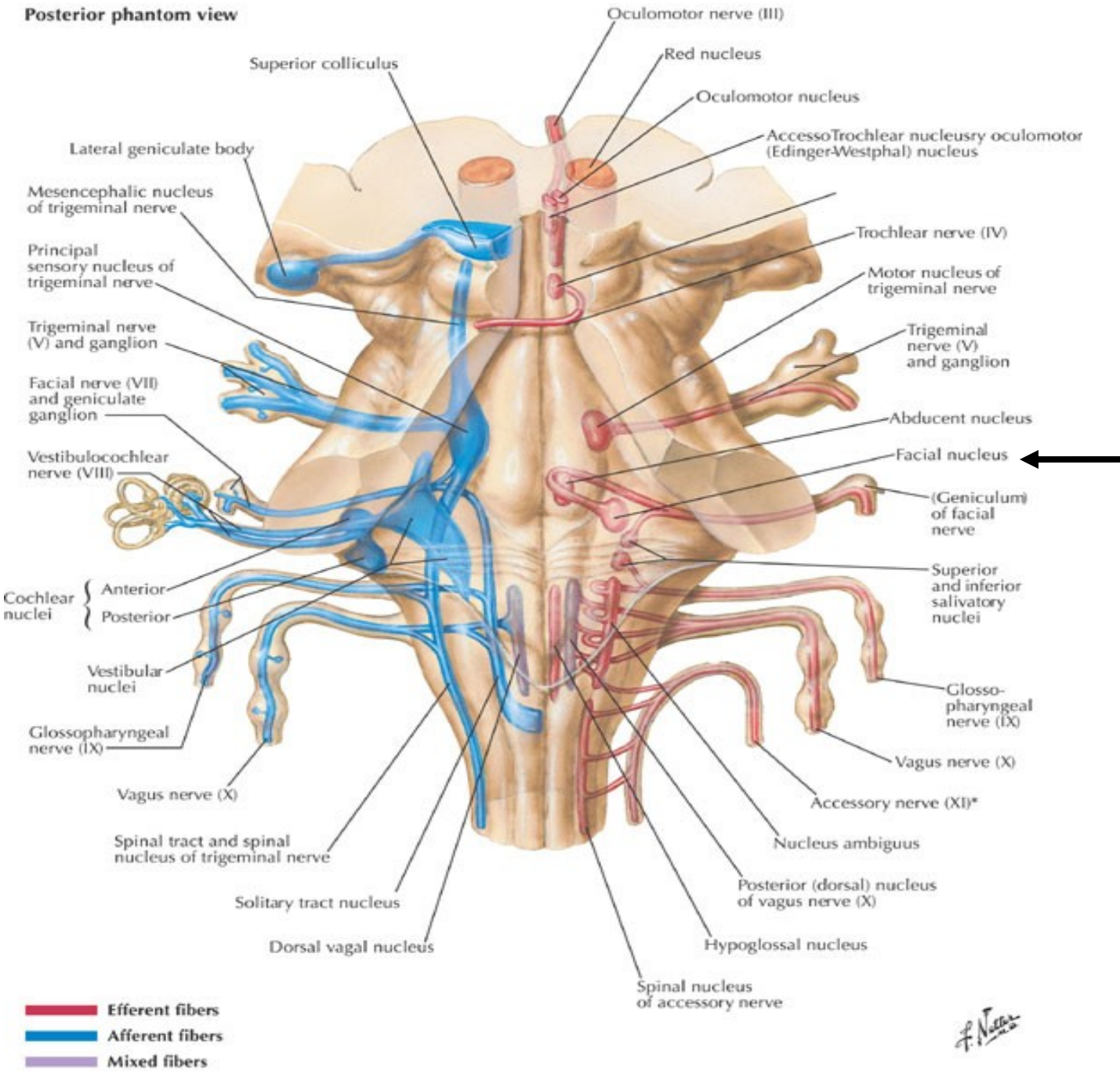
- The facial nerve consists of two parts.
- The **facial nerve proper** "motor root" consist **SVE (branchiomotor) neurons**
- The **nervus intermedius** (*pars intermedii of Wrisberg)* is sometimes referred to as the "sensory root," which is a misnomer since in addition to sensory fibers it also carries parasympathetic fibers like **GVE (secretomotor) parasympathetic neurons**, SVA (taste) sensation, GSA sensation from the area posterior to the ear, GVA sensation from the nasal cavity and soft palate.

NUCLEI

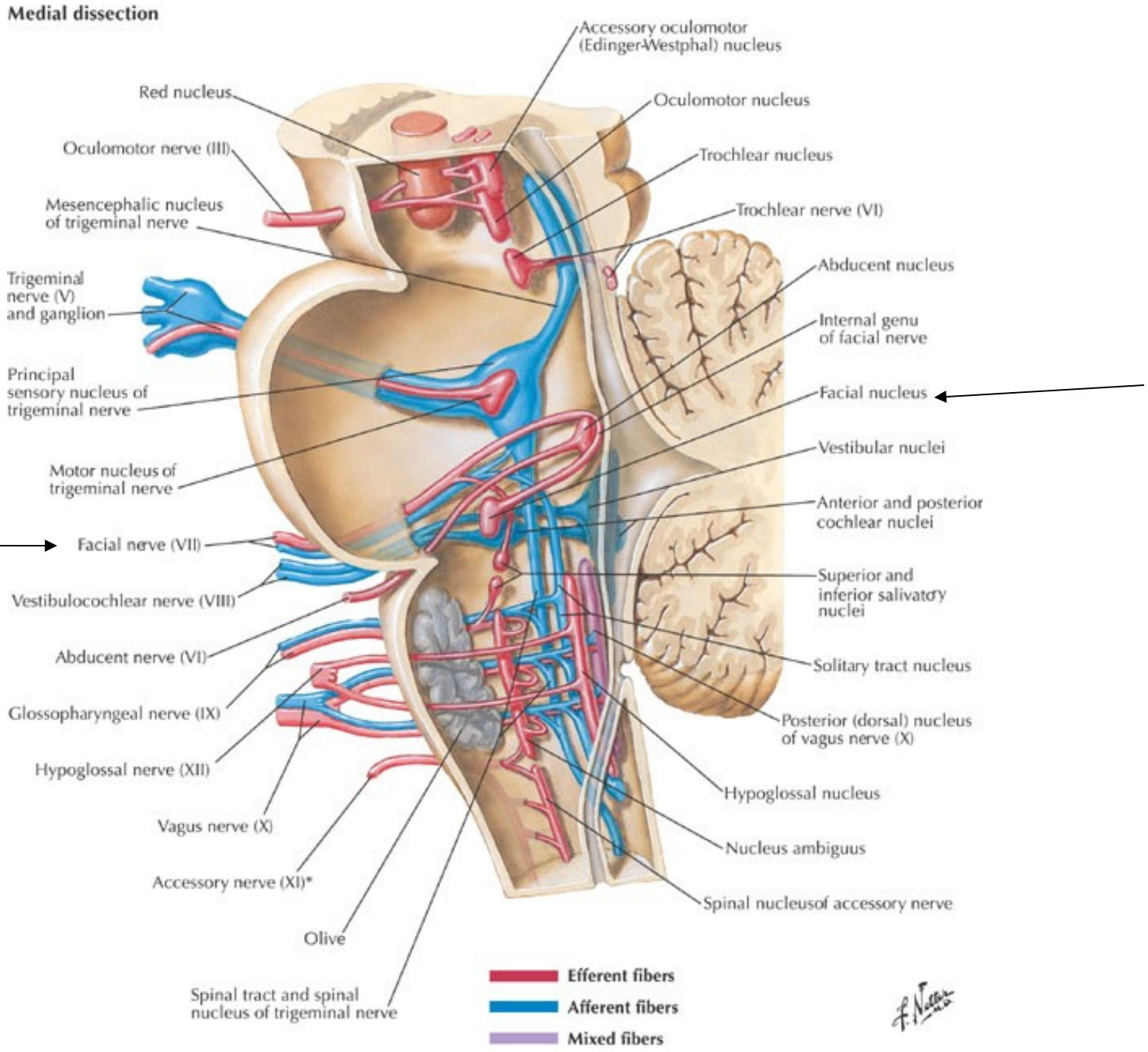
- Facial nerve consists of two motor and two sensory nuclei which lie in brain stem.



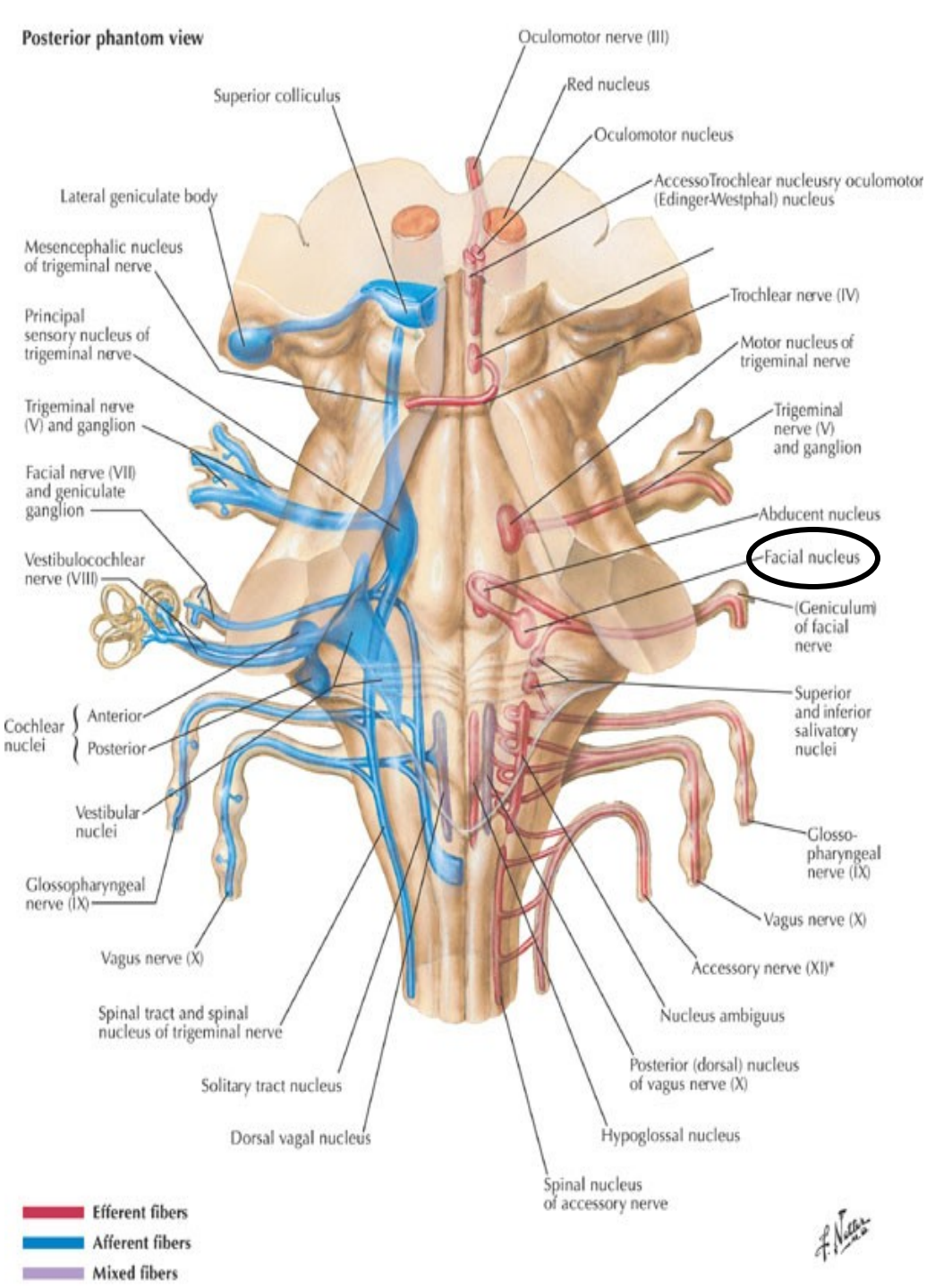
Facial nerve nuclei it is a Main motor nucleus lie deep in the reticular formation of lower part of pons.



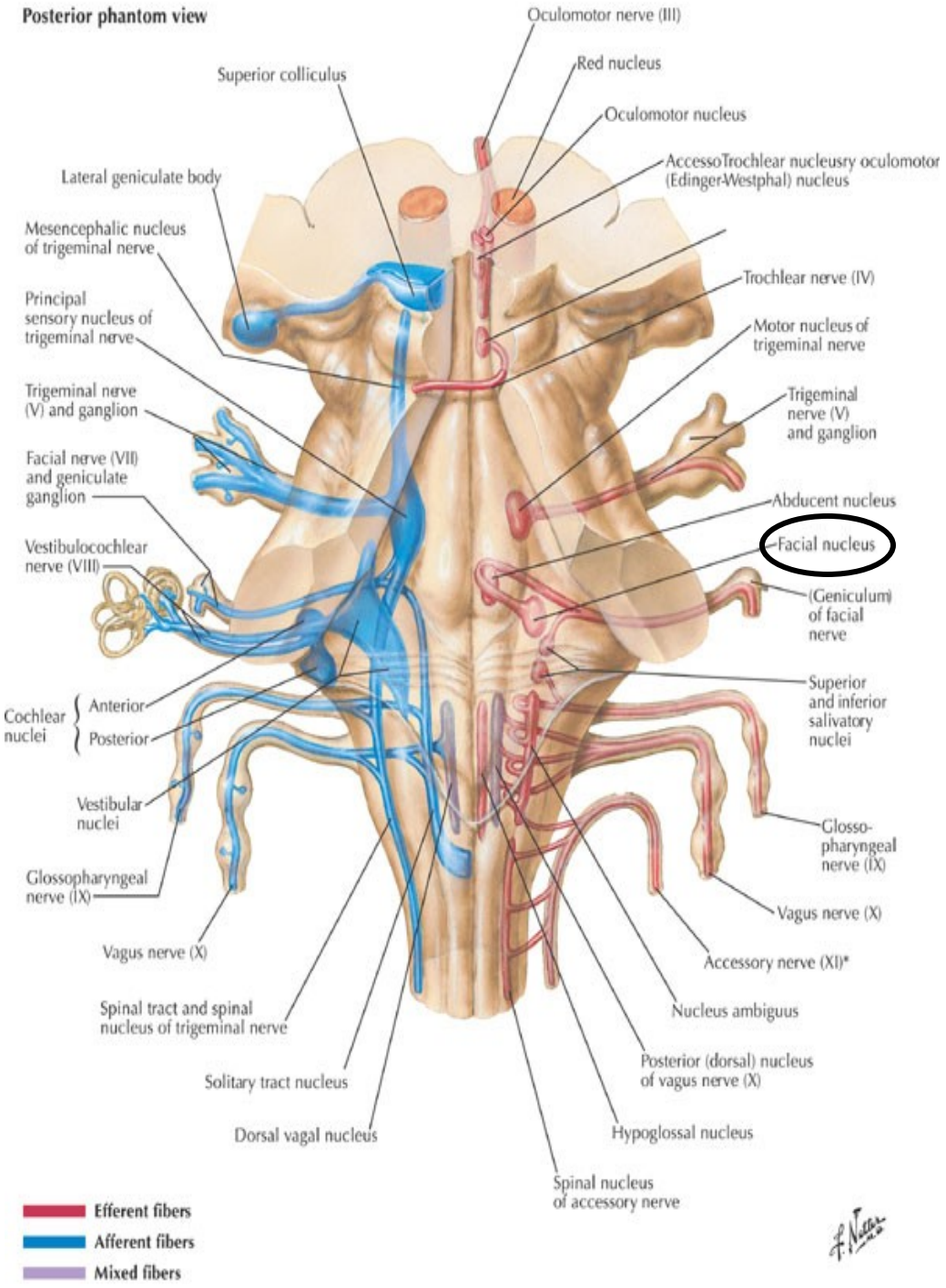
Cranial Nerve Nuclei in Brainstem: Schema



- The motor part of the facial nerve arises from the facial nerve nucleus.
- The fibers of the motor root first travel posteriorly around the medial side of the abducent nucleus. They then pass around the nucleus beneath the colliculus facialis in the floor of the fourth ventricle and finally, pass anteriorly to emerge from the brainstem

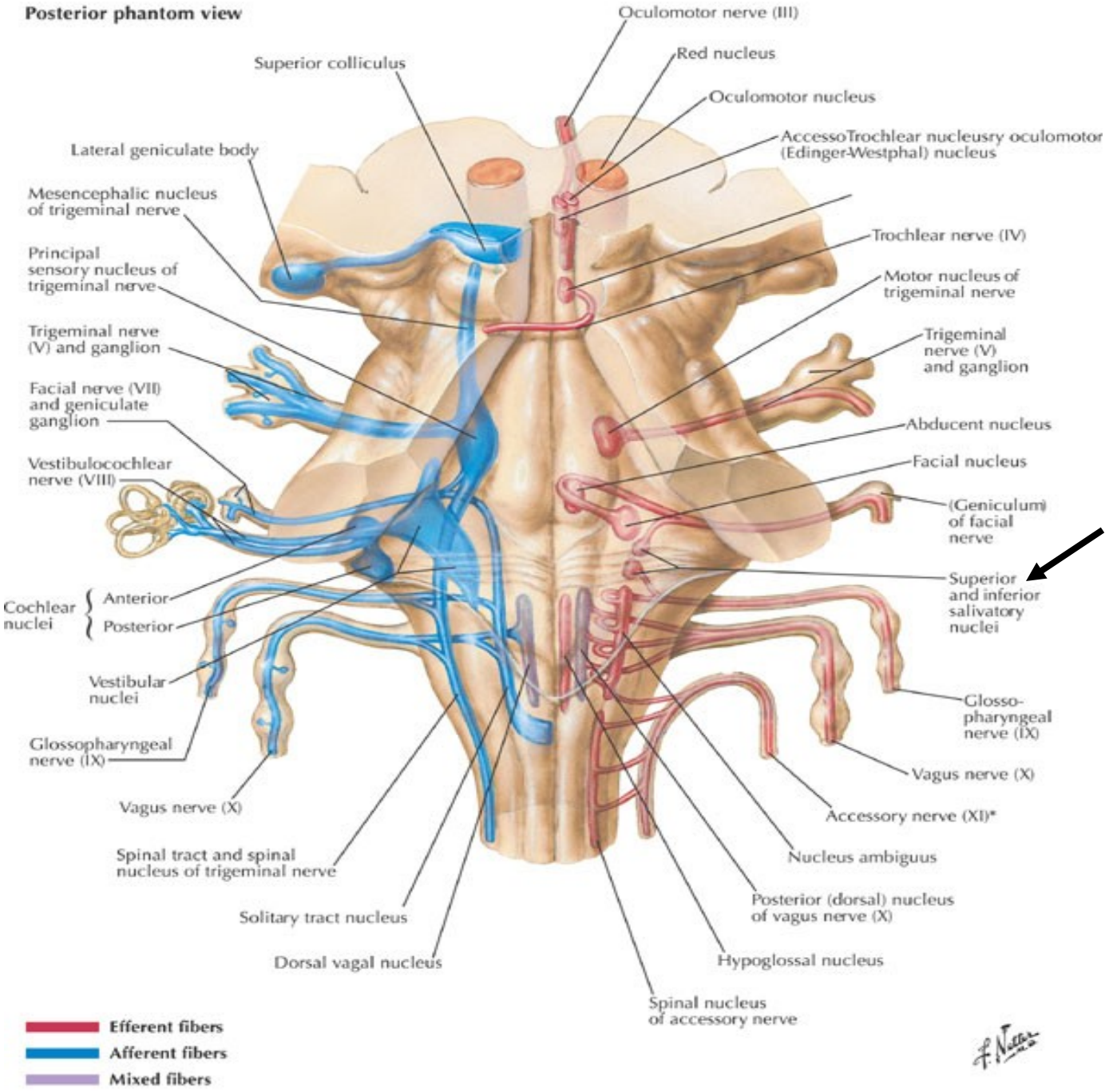


- The motor part supplies muscles derived from 2nd pharyngeal arch
- Somatic motor fibers to the muscles of the face, scalp, and auricle, the Buccinator, Platysma, the stapedius, the Stylohyoid, and posterior belly of the Digastric.

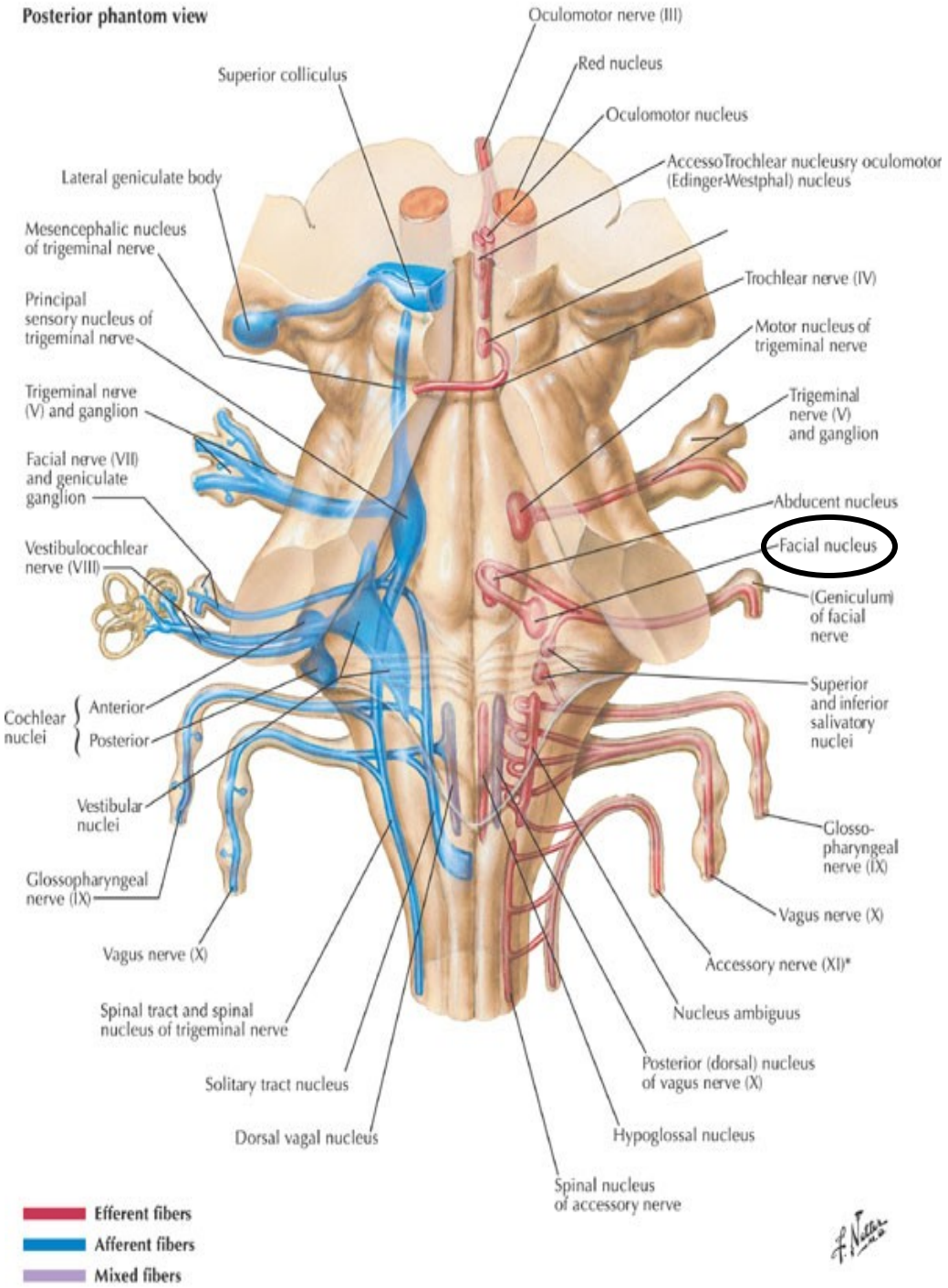


2) **Superior salivatory nucleus** is also motor nucleus

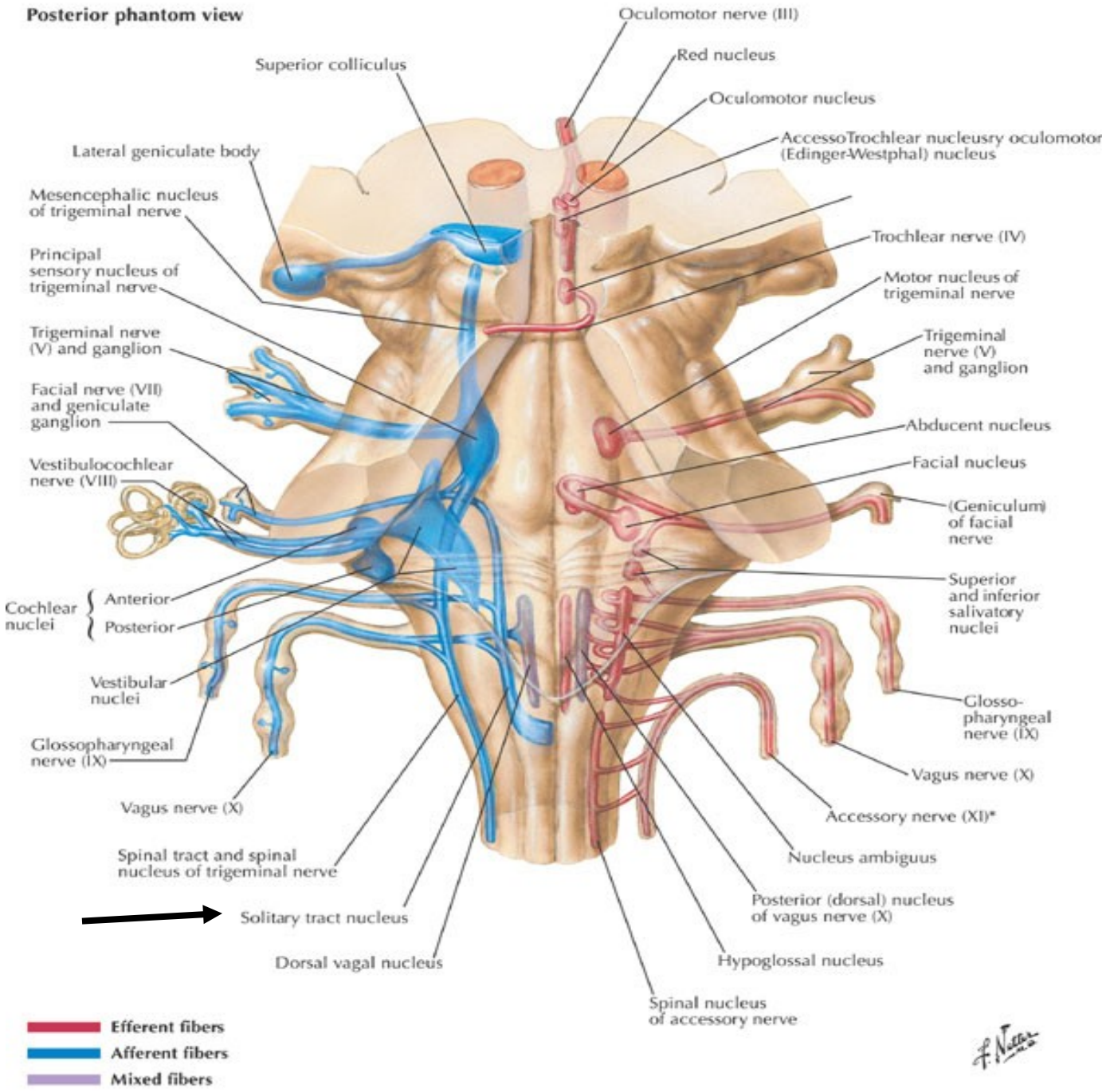
(Parasympathetic nuclei) lie posterolateral to main motor nucleus in pons contains GVE preganglionic parasympathetic nerve cell bodies whose axons leave the brainstem via the nervus intermedius. These preganglionic fibers are



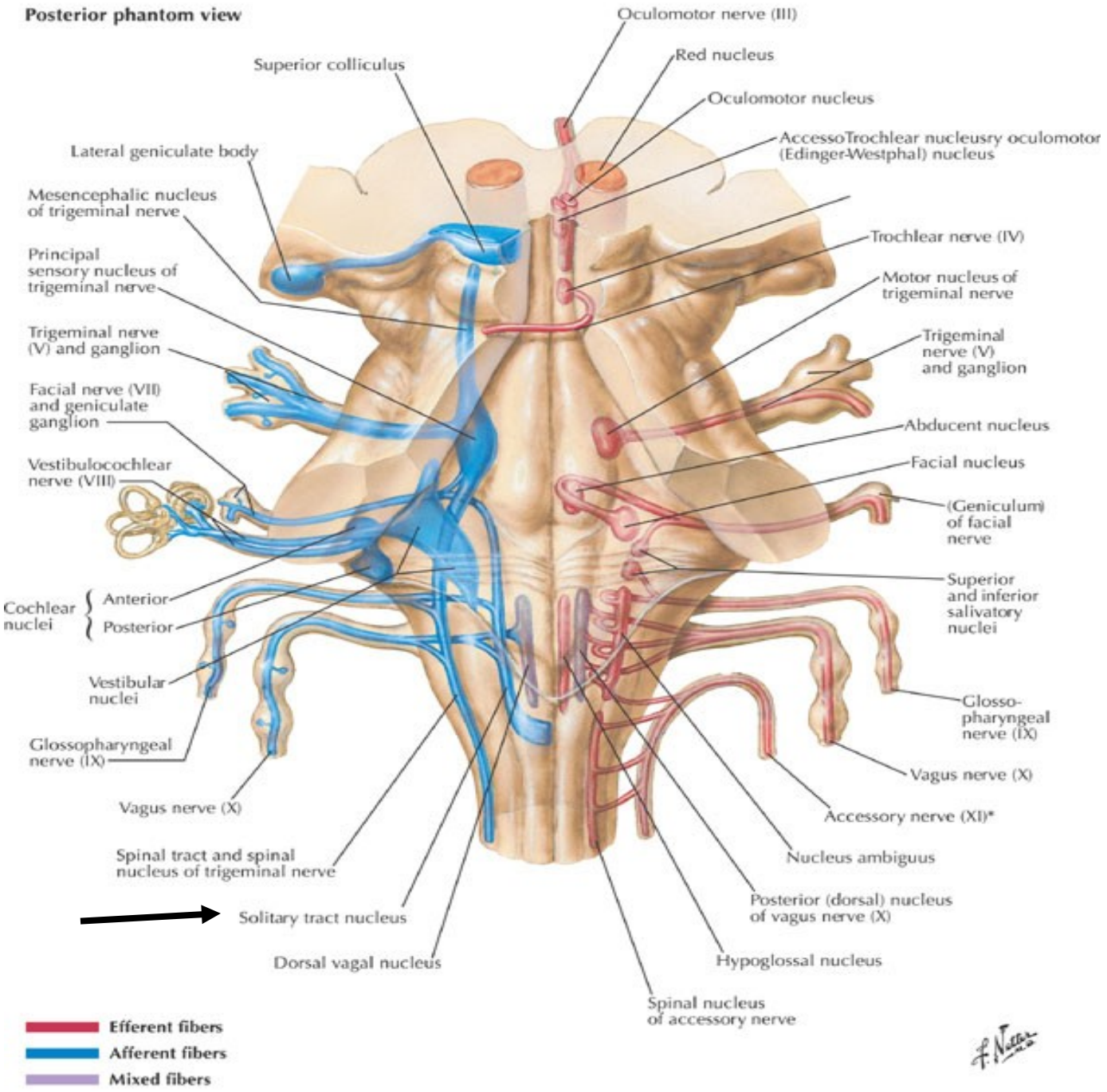
- Send sympathetic secretomotor fibers which constitute the vasodilator nerves to pterygopalatine & submandibular ganglion, mainly for lacrimal & salivary secretion.



3) Sensory nucleus is **solitary tract nucleus** lie close to motor nucleus in the upper medulla.

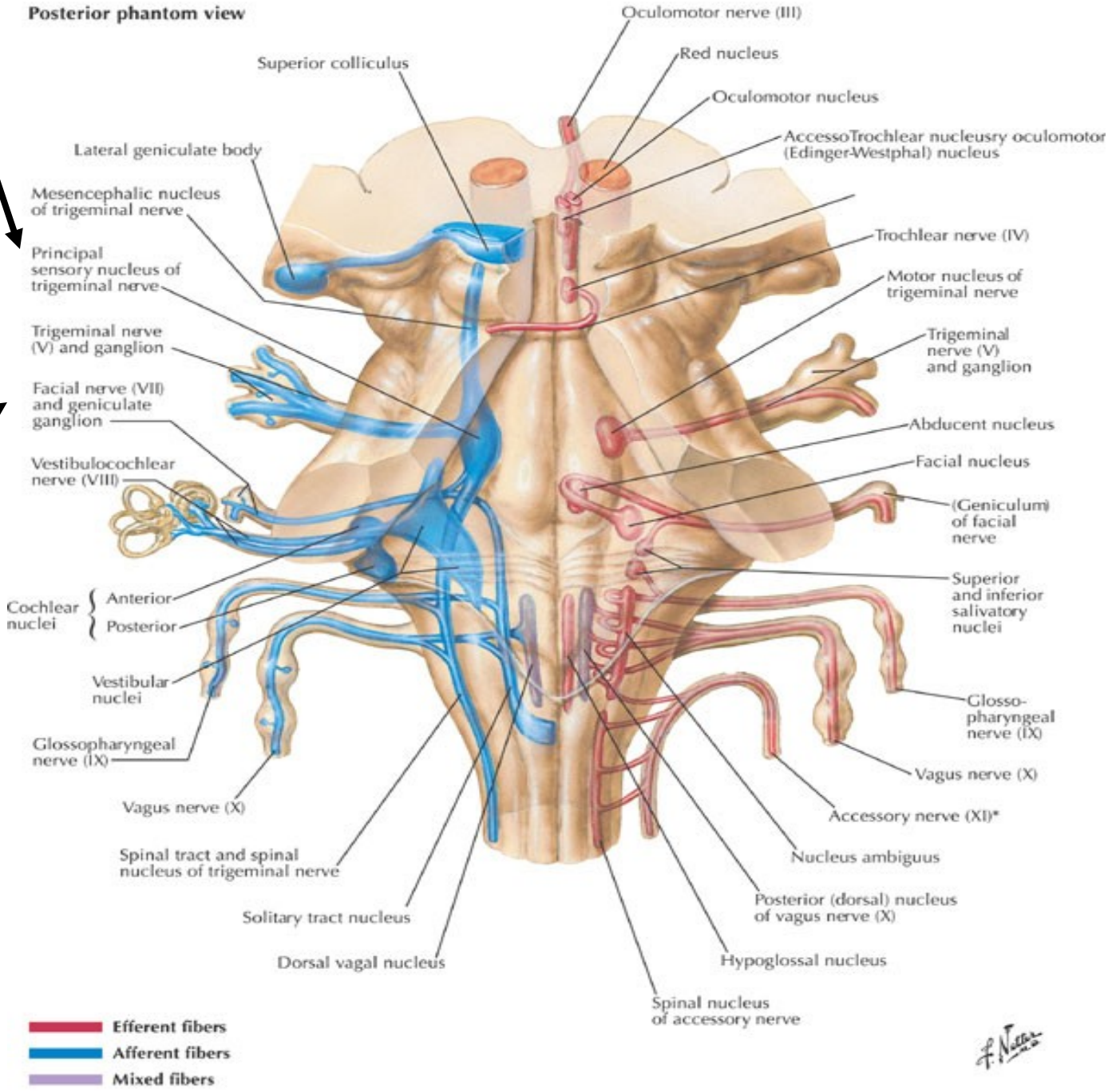


Recieves taste fibers of chorda tympani (from tongue)& of great petrosal nerve (from taste buds of soft palate) through geniculate ganglion.

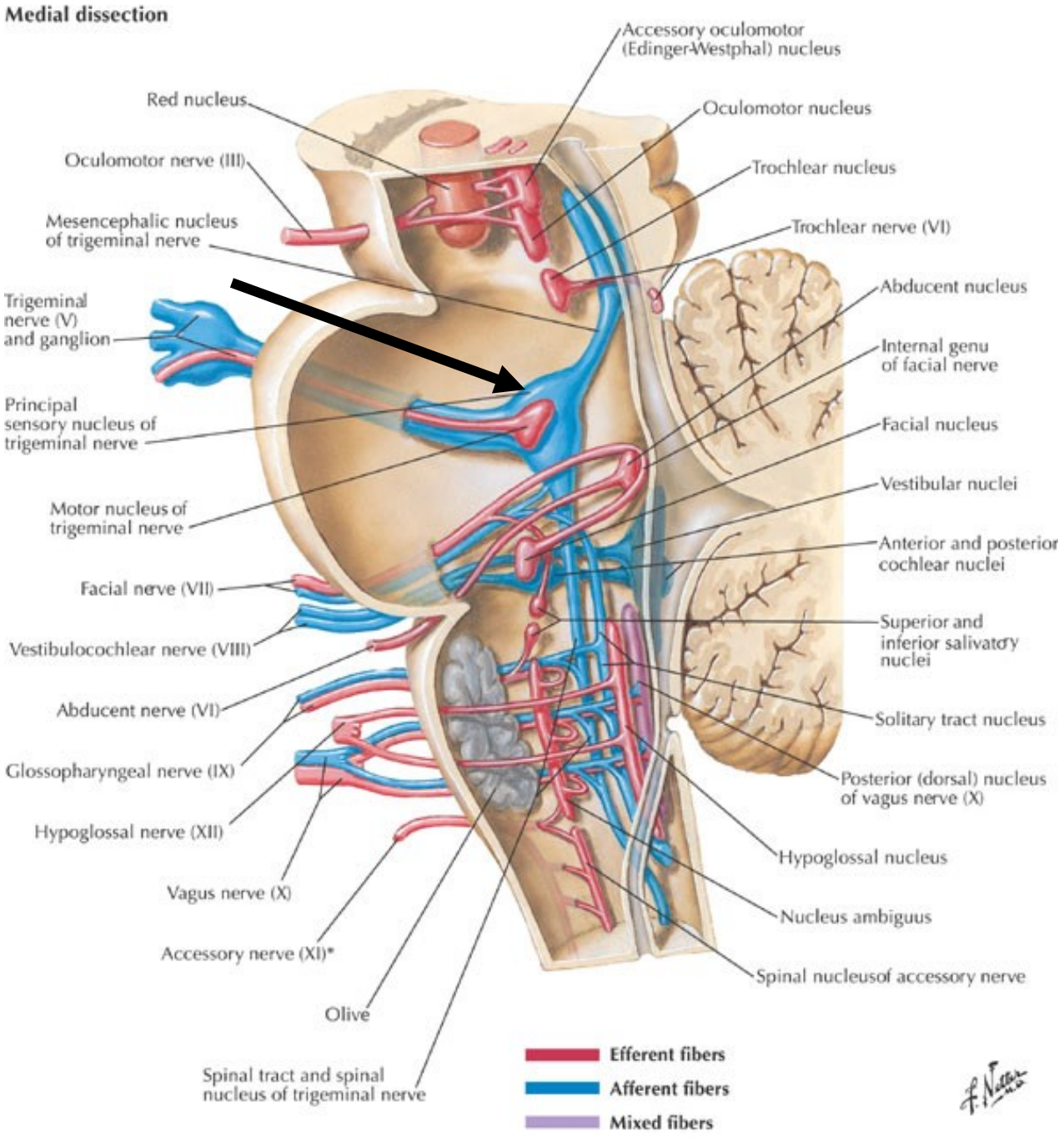


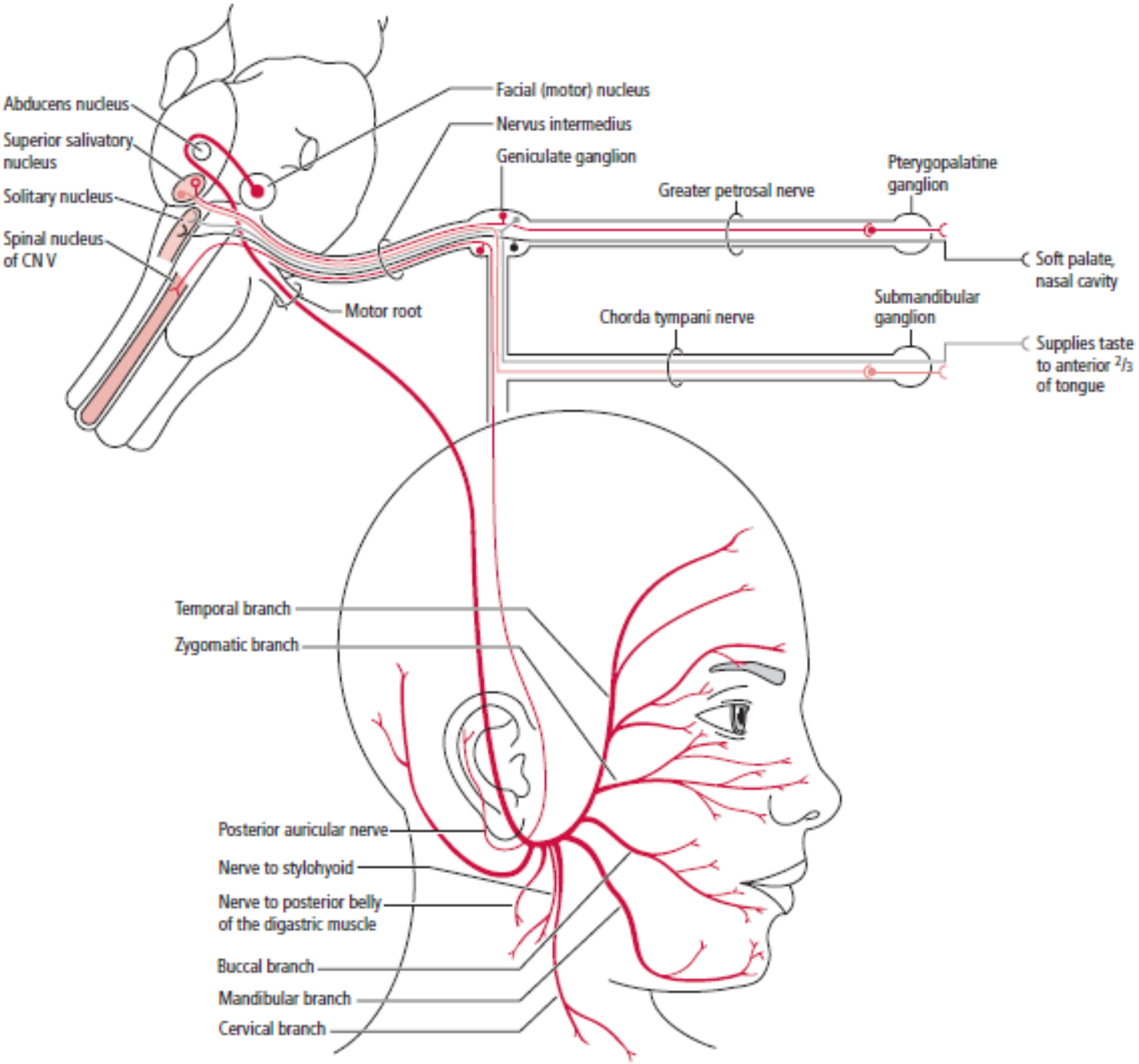
4) Sensory nuclei of trigerminal nerve.

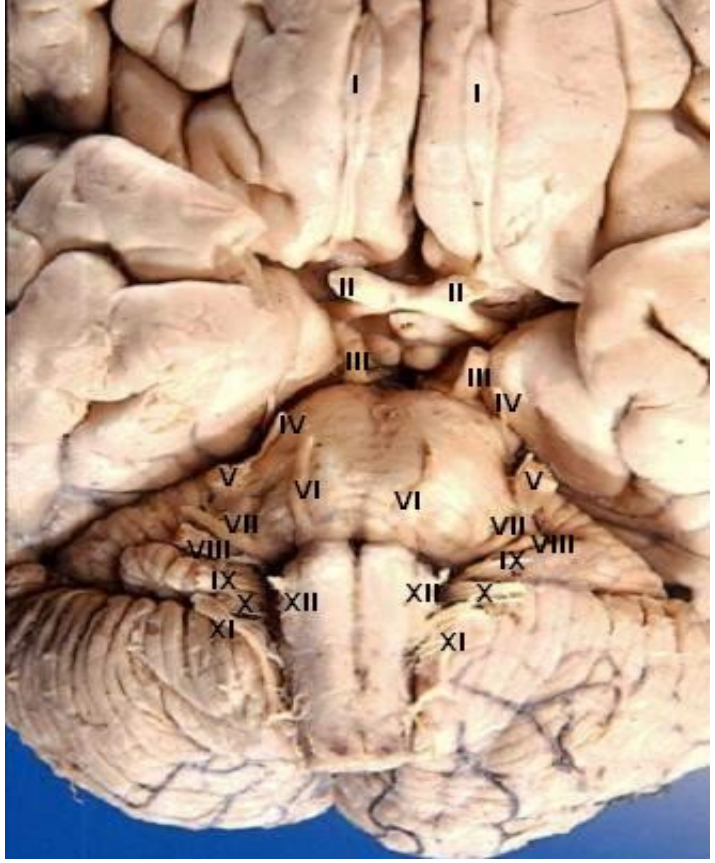
Supply skin of exernal acoustic meatus & tympanic membrane via geniculate ganglion.

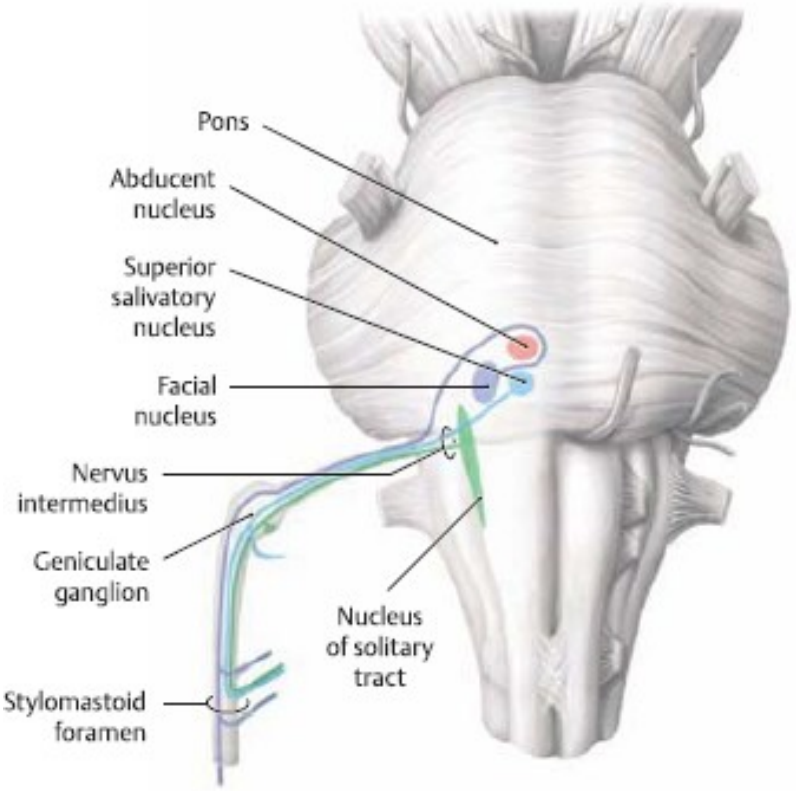


Cranial Nerve Nuclei in Brainstem: Schema



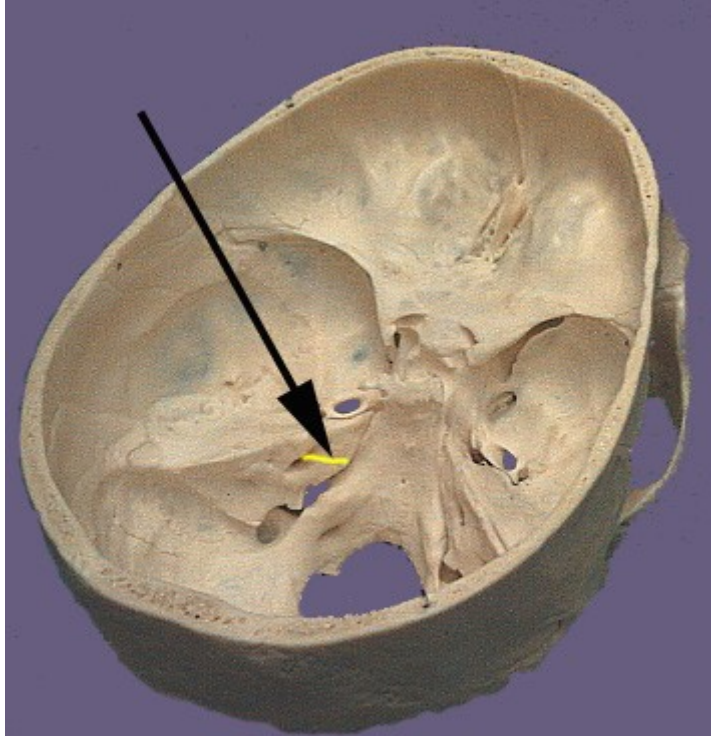

Abducens nucleus
Superior salivatory nucleus
Solitary nucleus
Spinal nucleus of CN V
Facial (motor) nucleus
Nervus intermedius
Geniculate ganglion
Motor root
Greater petrosal nerve
Pterygopalatine ganglion
Soft palate, nasal cavity
Chorda tympani nerve
Submandibular ganglion
Supplies taste to anterior 2/3 of tongue
Temporal branch
Zygomatic branch
Posterior auricular nerve
Nerve to stylohyoid
Nerve to posterior belly of the digastric muscle
Buccal branch
Mandibular branch
Cervical branch

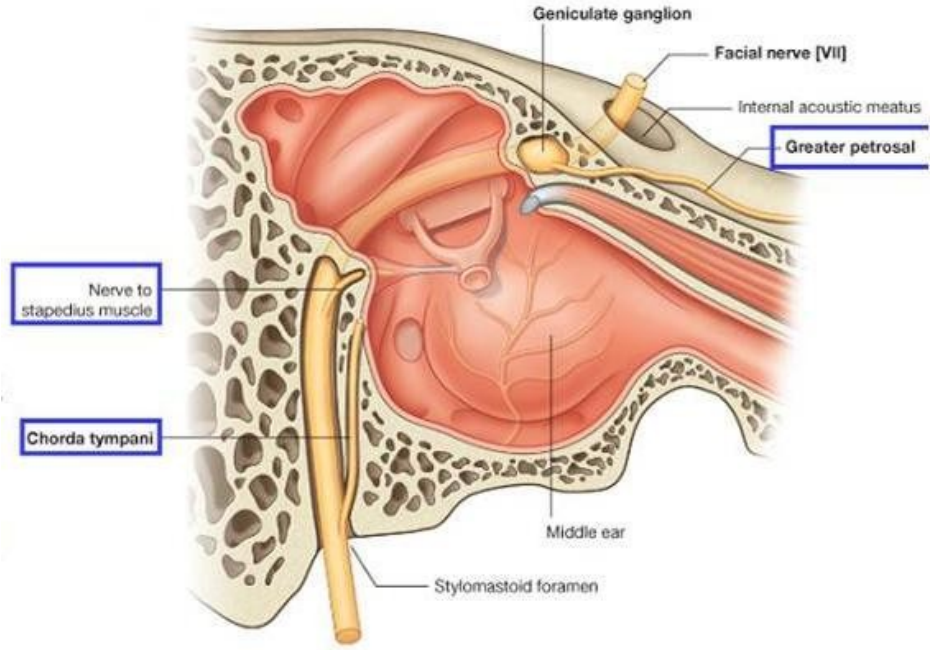


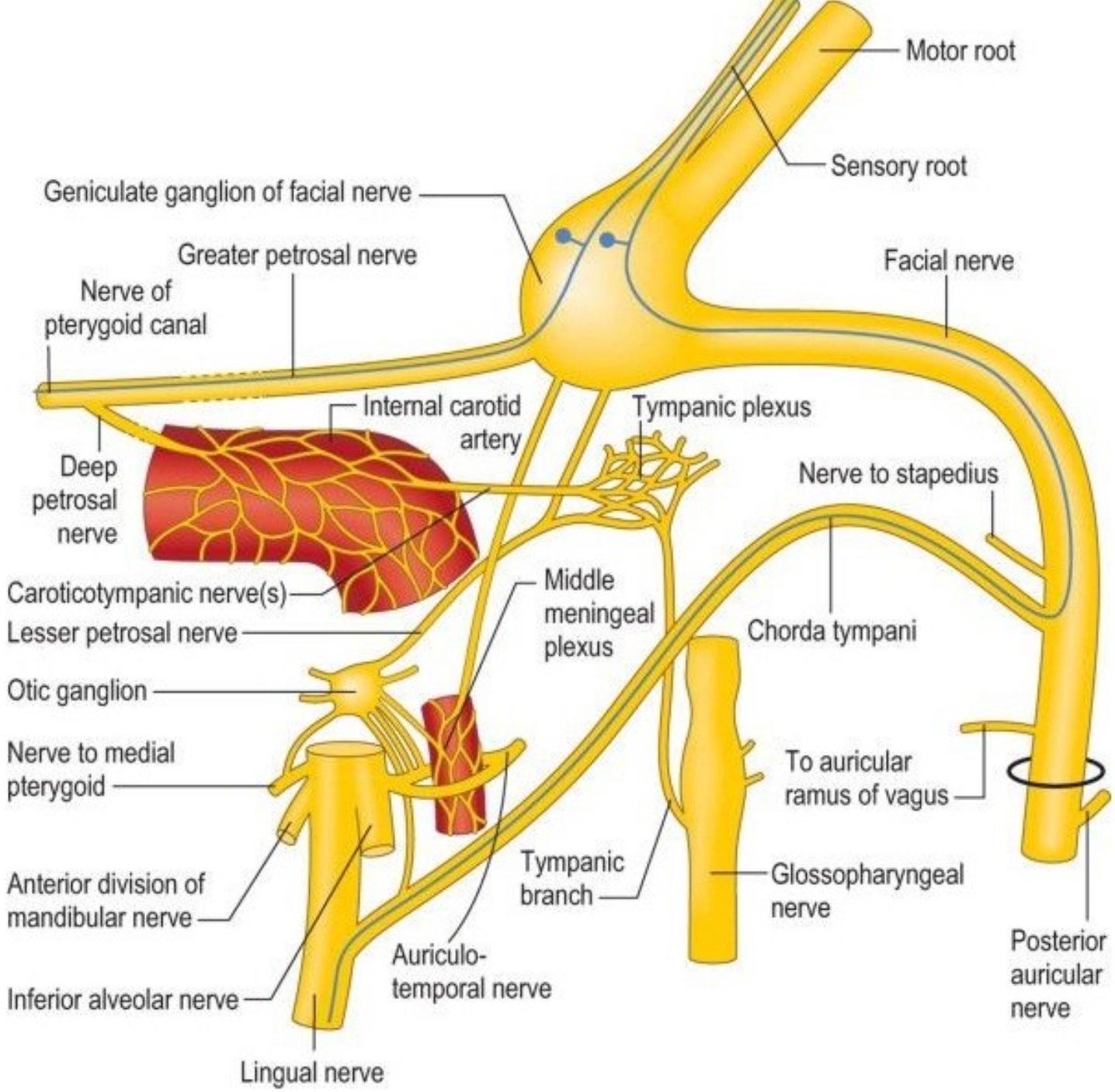


- Both nerve roots (motor root and nervus intermedius) facial (vii) emerge **from the brainstem** at the **cerebellopontine angle (**groove between the pons and the medulla oblongata)

- Near their exit from the brainstem, the two roots of the facial nerve accompany one another to the **internal acoustic meatus** of the **petrous portion of the temporal bone** accompanied by the vestibulocochlear nerve and proceed to the **facial canal** in the petrous temporal bone where the nervus intermedius presents a swelling—the **geniculate ganglion** which contains the cell bodies of all the sensory fibers .



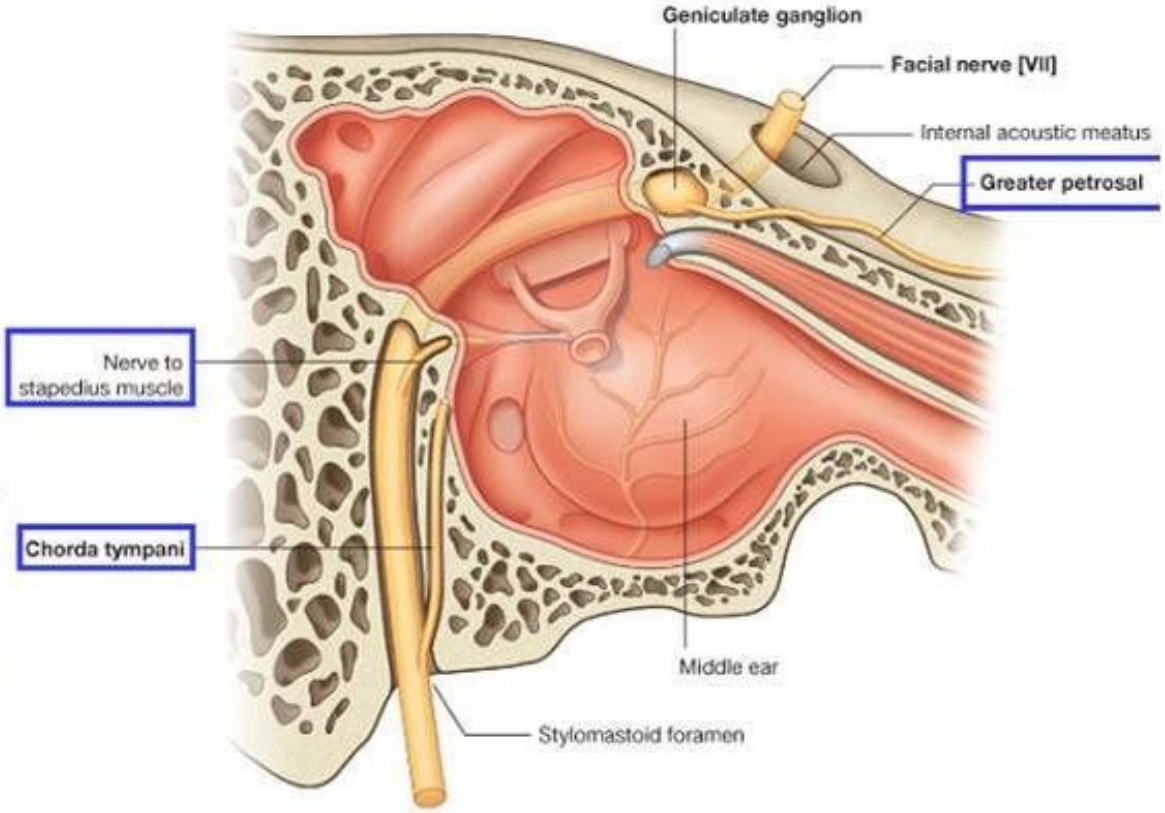



Motor root
Sensory root
Geniculate ganglion of facial nerve
Greater petrosal nerve
Facial nerve
Nerve of pterygoid canal
Internal carotid artery
Tympanic plexus
Deep petrosal nerve
Nerve to stapedius
Caroticotympanic nerve(s)
Middle meningeal plexus
Lesser petrosal nerve
Chorda tympani
Otic ganglion
Nerve to medial pterygoid
To auricular ramus of vagus
Tympanic branch
Glossopharyngeal nerve
Anterior division of mandibular nerve
Auriculo-temporal nerve
Posterior auricular nerve
Inferior alveolar nerve
Lingual nerve

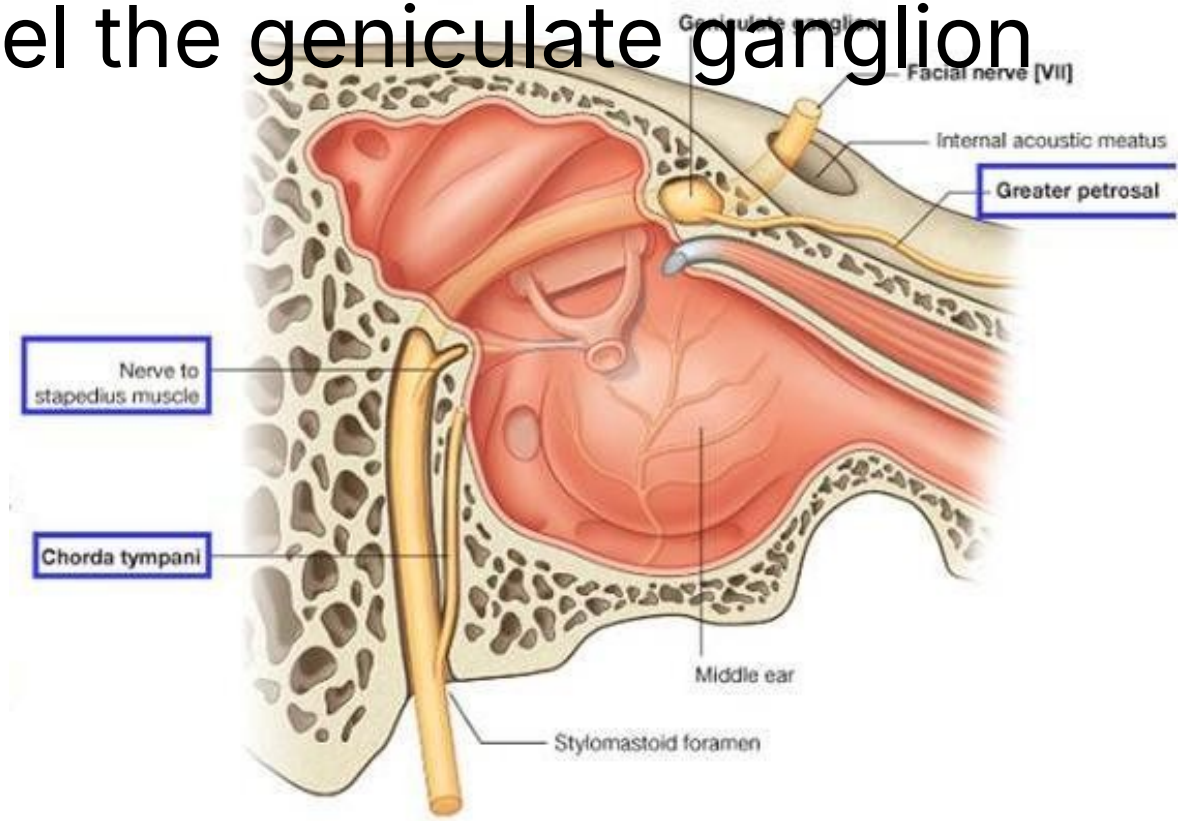
facial canal

- The facial nerve gives rise to three of its branches in the facial canal: the greater petrosal nerve, the nerve to the stapedius muscle (which innervates the stapedius muscle in the middle ear), and the chorda tympani nerve.

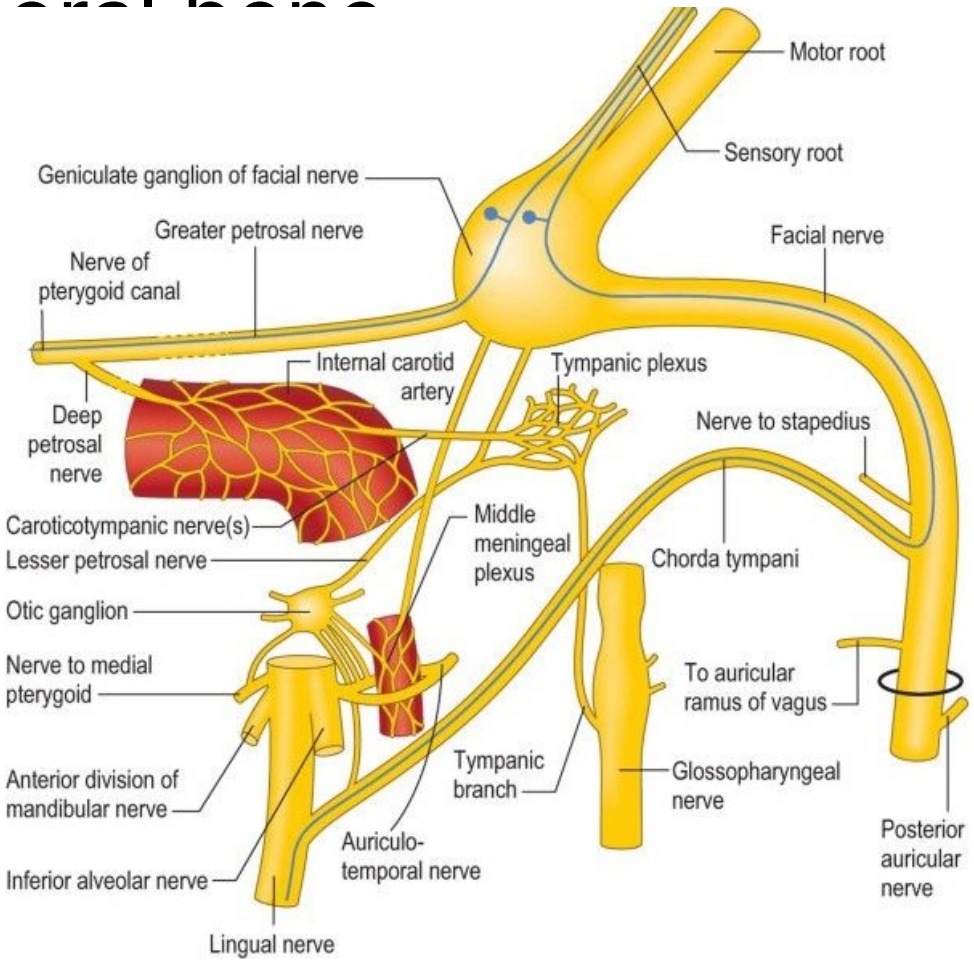
greater petrosal nerve

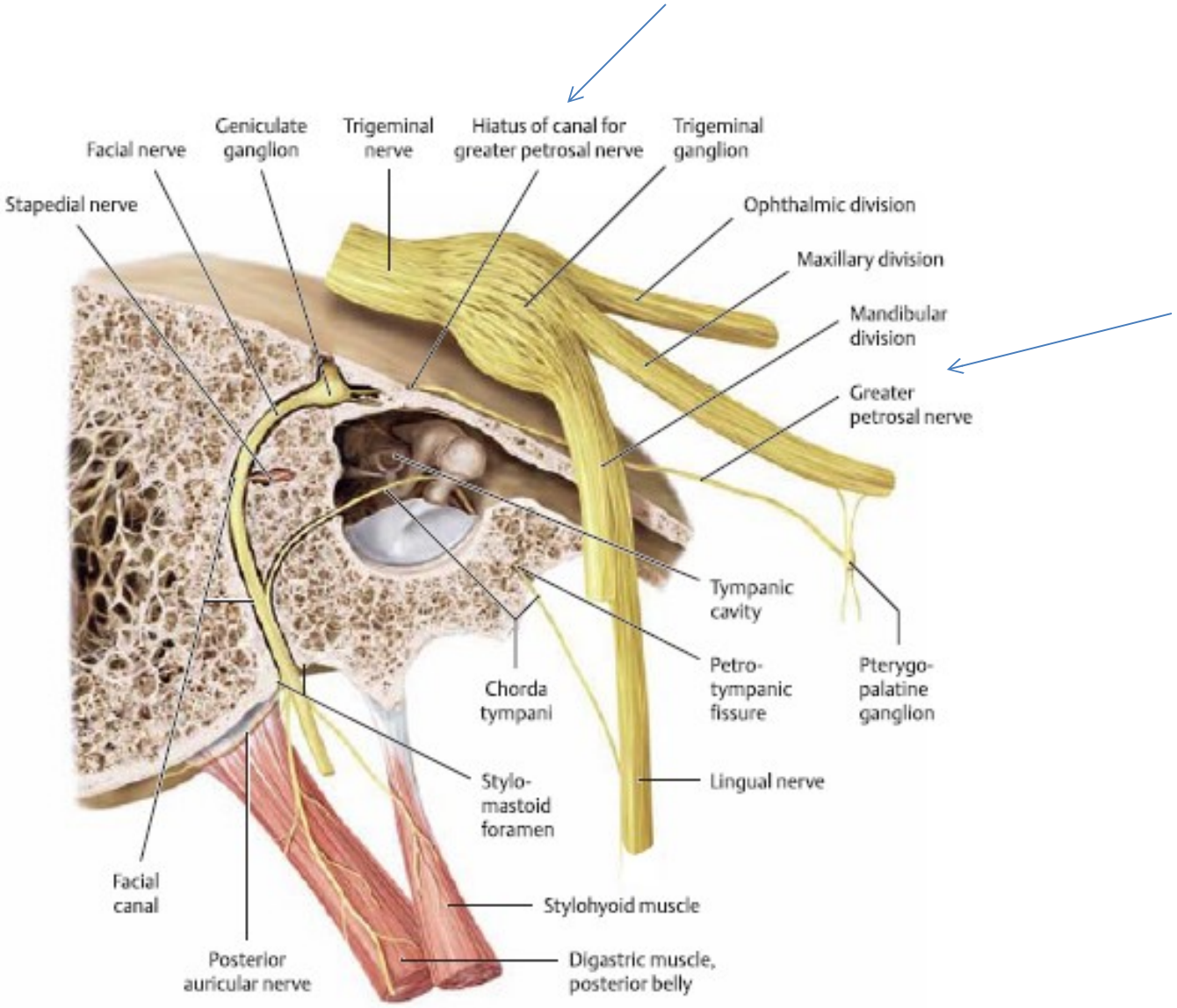


- The greater petrosal nerve, a branch of the nervus intermedius containing parasympathetic fibres destined for the pterygopalatine ganglion, and taste fibres from the palate, leaves the facial nerve at the level the geniculate ganglion



- The greater petrosal nerve passes anteriorly, receives a branch from the tympanic plexus and traverses a hiatus on the anterior surface of the petrous part of the temp[illegible]




Facial nerve
Geniculate ganglion
Trigeminal nerve
Hiatus of canal for greater petrosal nerve
Trigeminal ganglion
Stapedial nerve
Ophthalmic division
Maxillary division
Mandibular division
Greater petrosal nerve
Tympanic cavity
Petro-tympanic fissure
Pterygo-palatine ganglion
Chorda tympani
Stylo-mastoid foramen
Lingual nerve
Facial canal
Stylohyoid muscle
Posterior auricular nerve
Digastric muscle, posterior belly

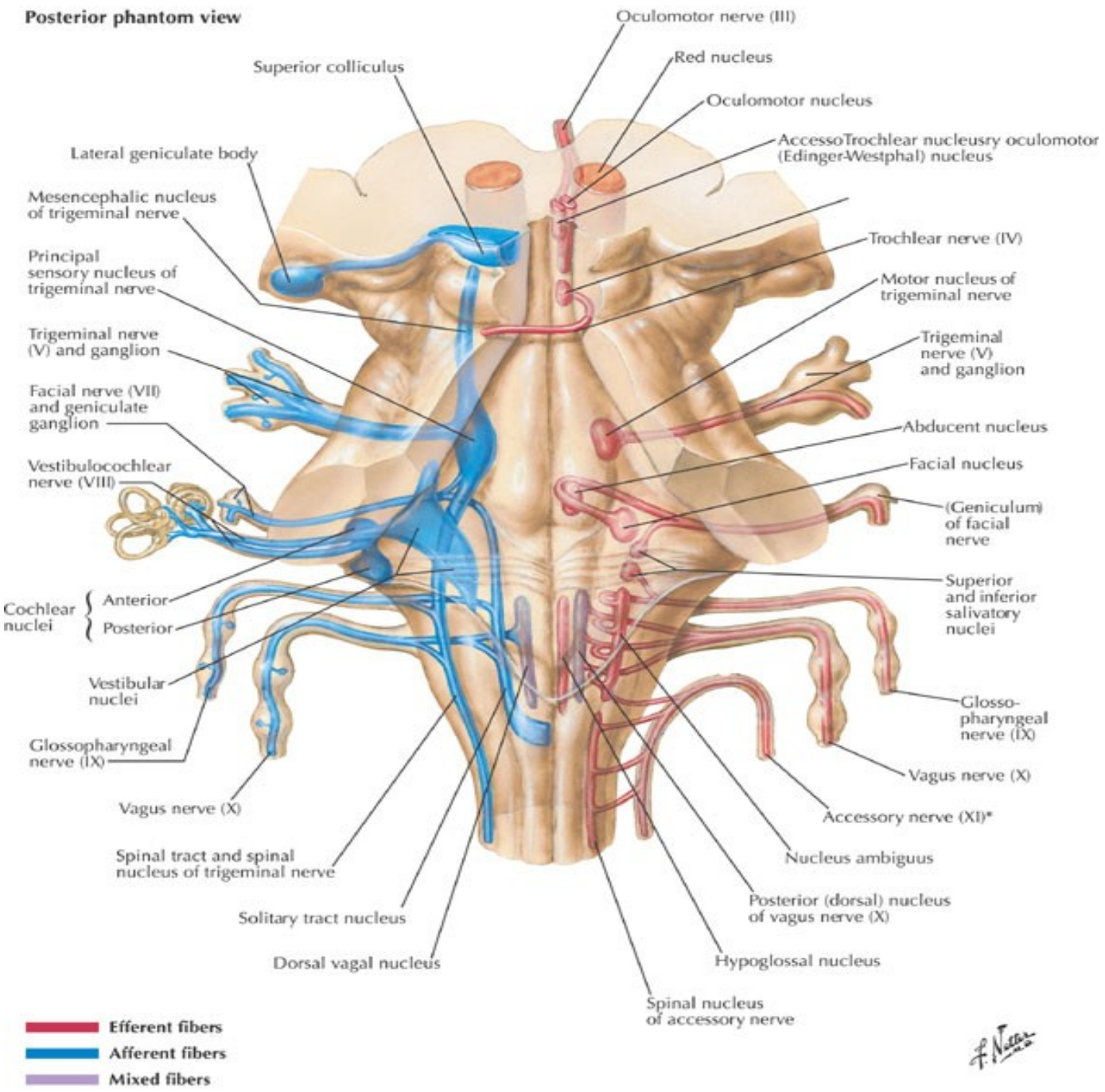
Facial nerve nuclei

1)Main motor nucleus lie deep in the reticular formation of lower part of pons.

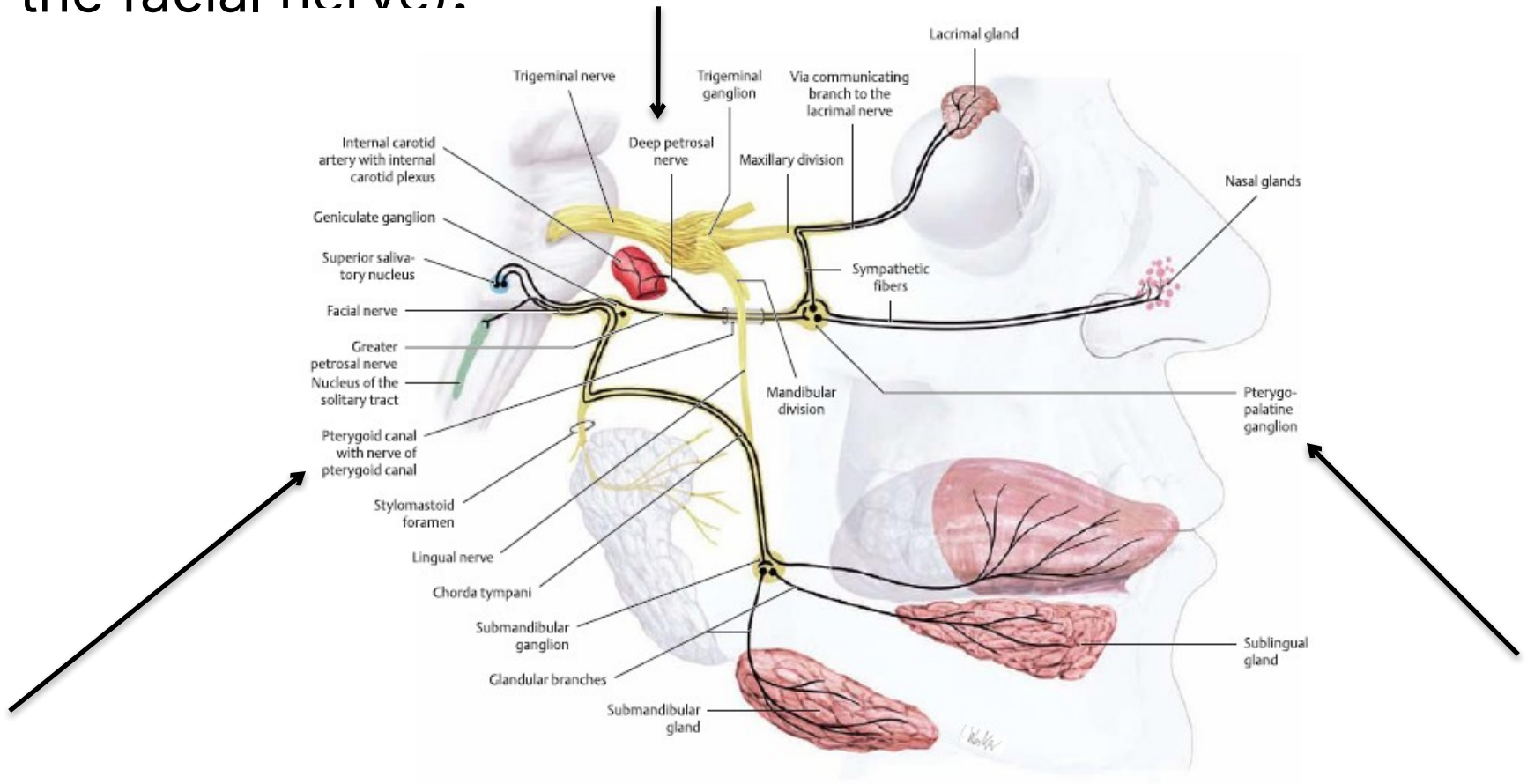
2)

Parasympathetic nuclei lie posterolateral to main motor nucleus (superior salivatory & lacrimal nuclei)

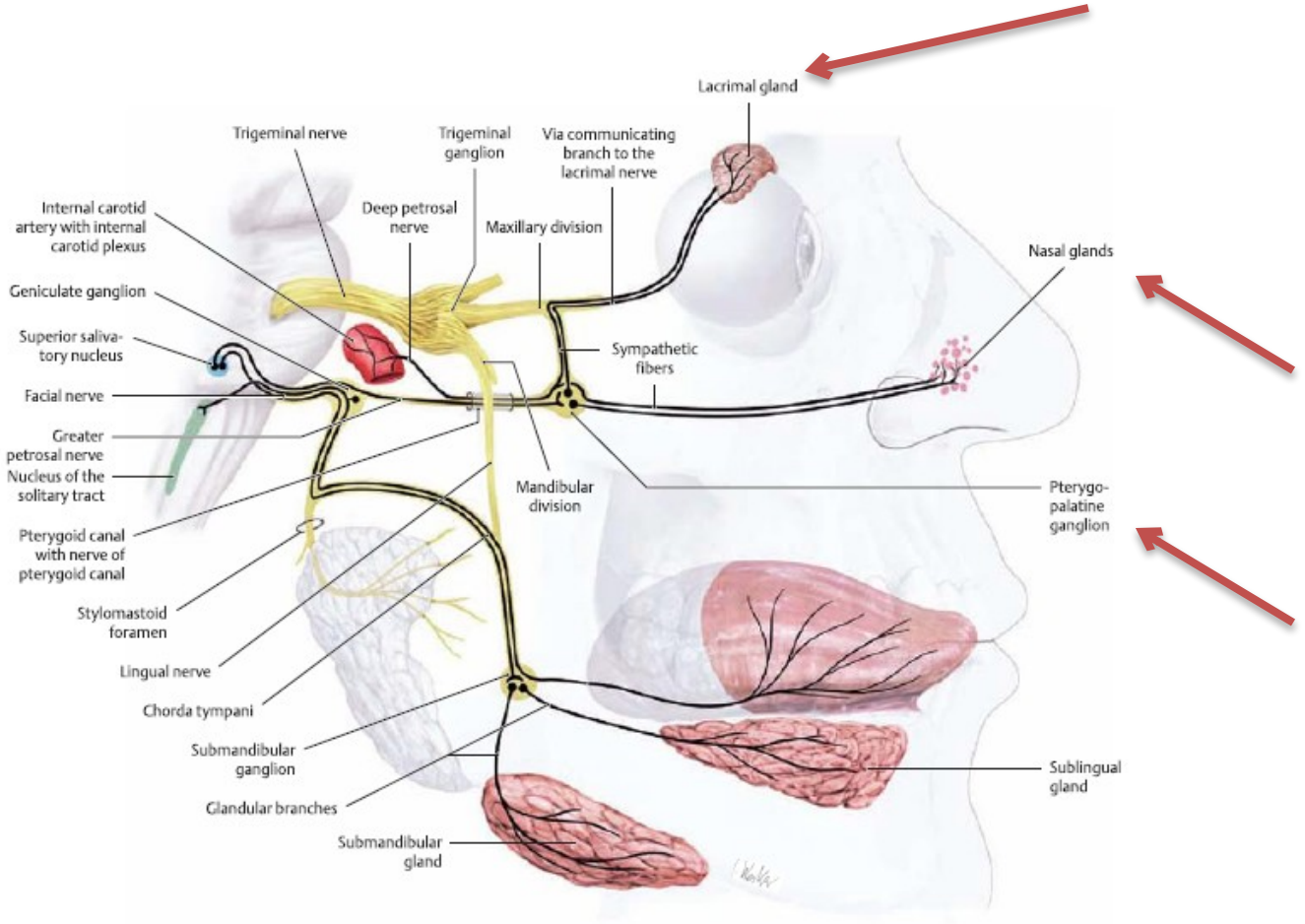
3) Sensory nucleus is upper part of tractus solitarius & lie close to motor nucleus.



- It enters the middle cranial fossa to reach the foramen lacerum. Here it is joined by the deep petrosal nerve from the internal carotid sympathetic plexus, to become the nerve of the pterygoid canal (Vidian nerve) to enter the pterygopalatine fossa where they terminate and synapse in the **pterygopalatine ganglion (a parasympathetic** ganglion of the facial nerve).

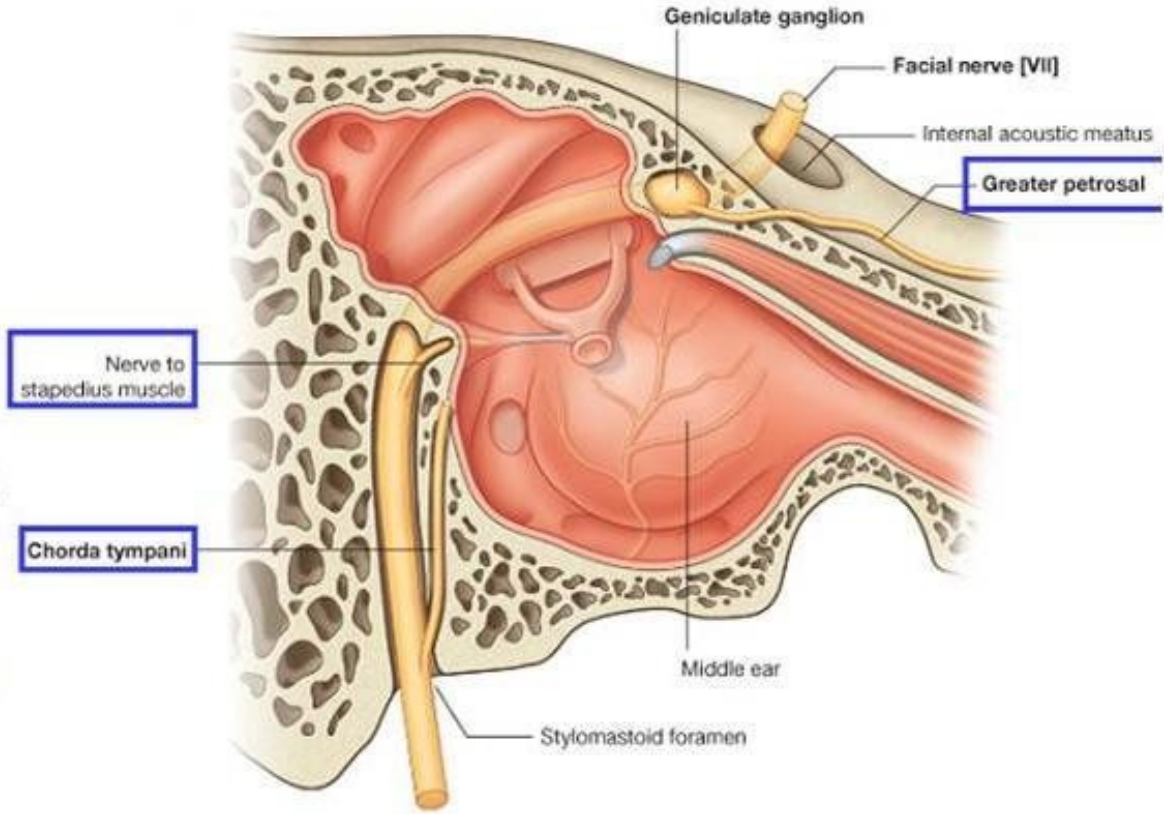


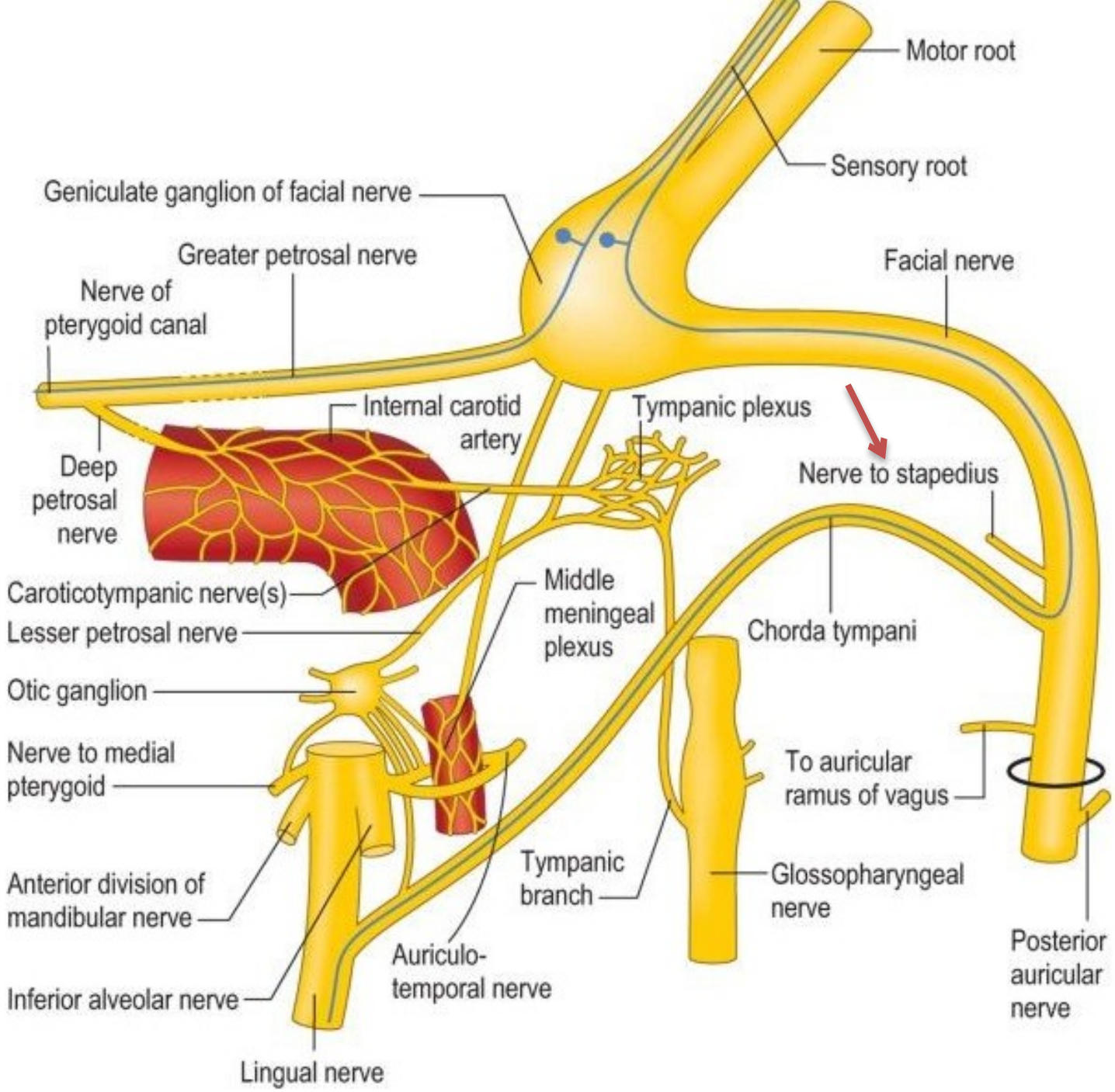
- Postganglionic parasympathetic fibers from the **lacrimal nucleus** leave the **pterygopalatine** ganglion and are distributing secretomotor innervation to the lacrimal gland via zygomatic and lacrimal branches of the maxillary nerve or pass directly from the ganglion, eliciting the production of tears and the glands of the nasal mucosa and oral cavity eliciting the production of mucus.



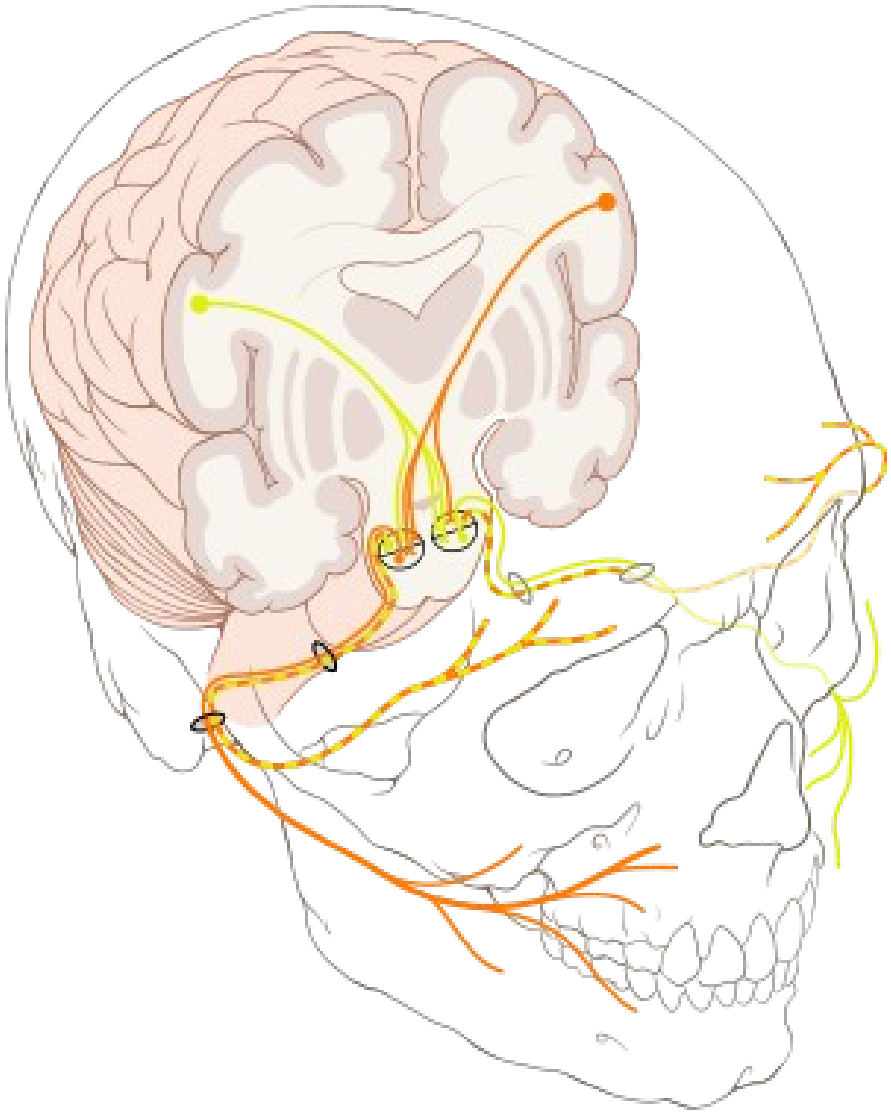
Nerve to the stapedius muscle

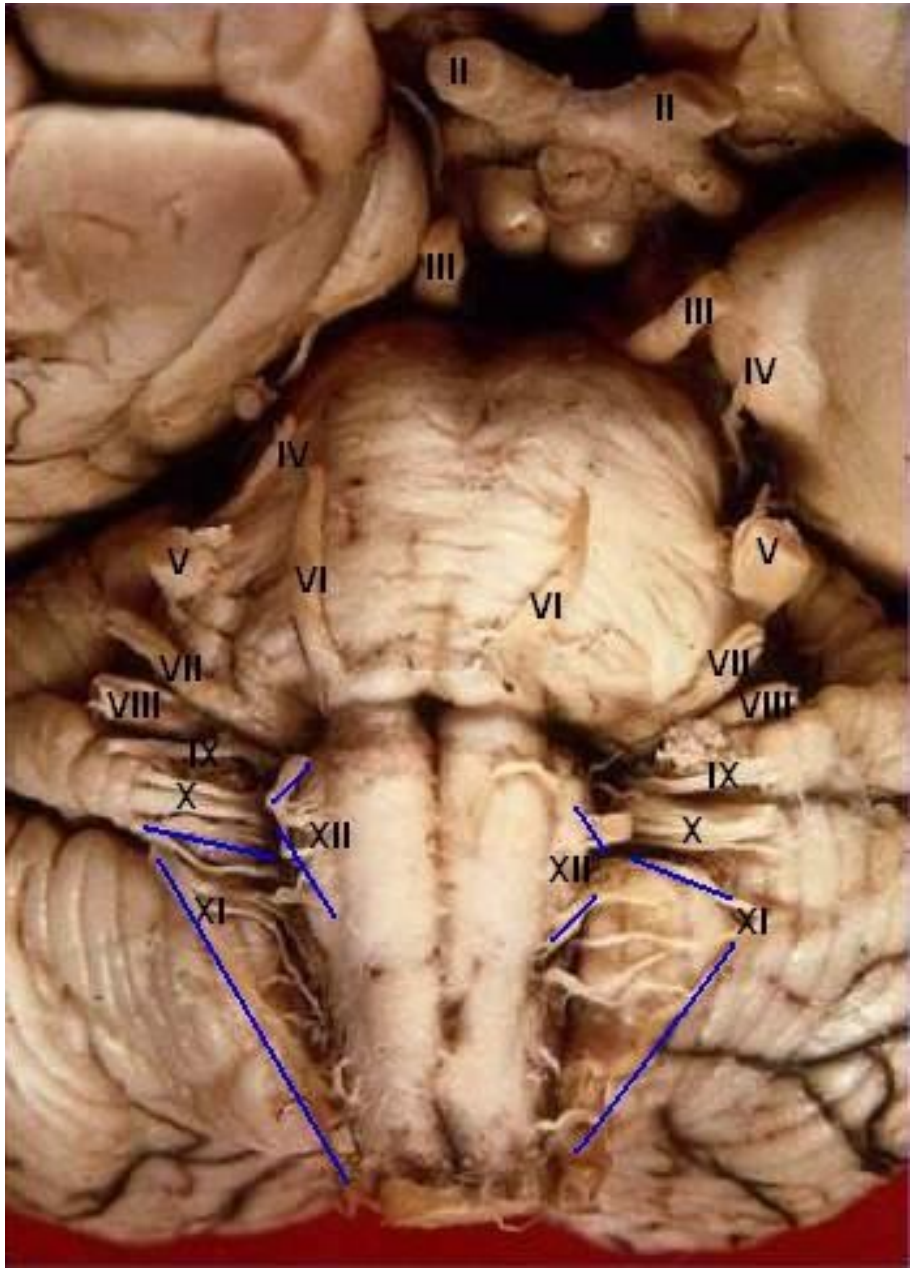
- Distal to the geniculate ganglion, the facial canal and its contained nerve bend sharply backward and downward, being now in the medial wall of the tympanic (middle ear) cavity, separated from the air by only the mucous membrane and a very thin layer of bone. Here **a motor branch goes to the stapedius muscle**

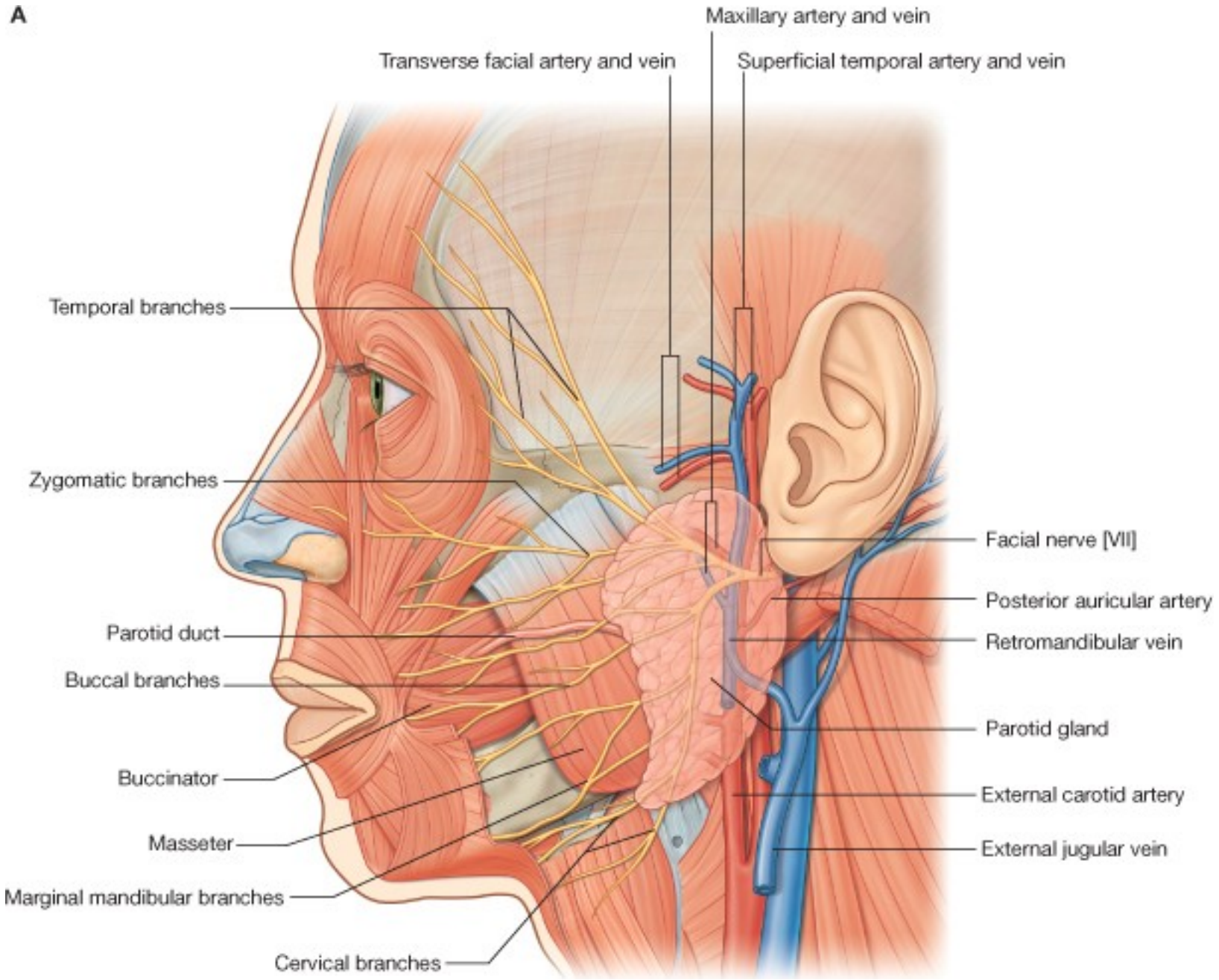



Motor root
Sensory root
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To auricular ramus of vagus
Tympanic branch
Glossopharyngeal nerve
Anterior division of mandibular nerve
Auriculo-temporal nerve
Posterior auricular nerve
Inferior alveolar nerve
Lingual nerve

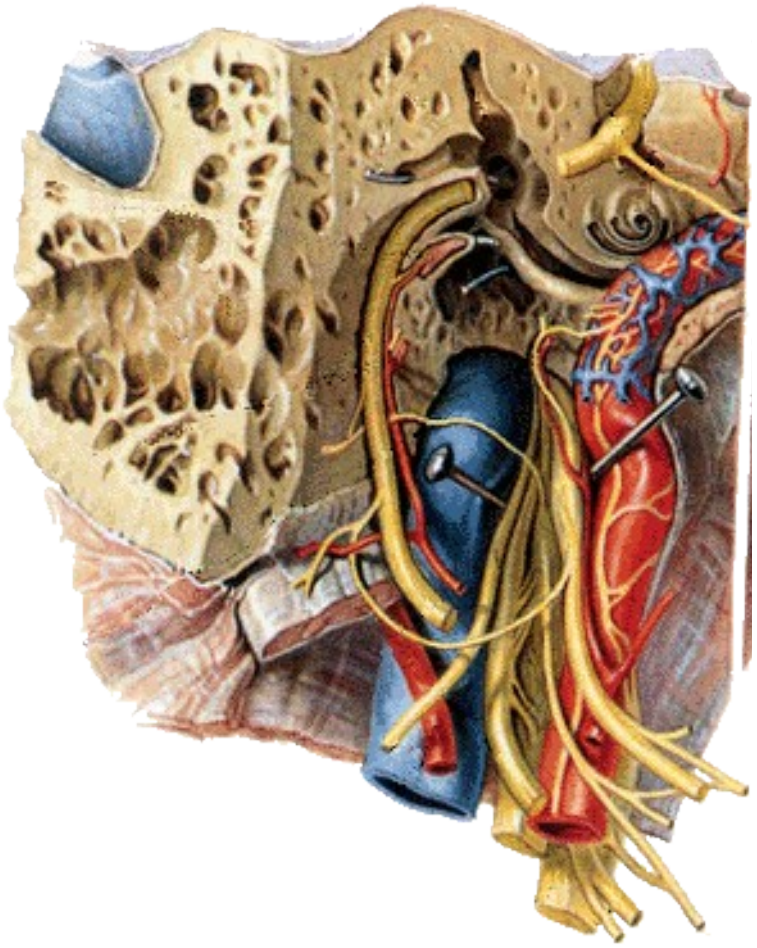
The pyramids end at the pontomedullary junction, noted most obviously by the large basal pons. From this junction facial nerve emerge.
The motor part of the facial nerve arises from the facial nerve nucleus in the pons while the sensory part of the facial nerve arises from the nervus intermedius.
The motor part and sensory part of the facial nerve enters the petrous temporal bone into the internal auditory meatus (intimately close to the inner ear) then runs a tortuous course (including two tight turns) through the facial canal, emerges from the stylomastoid foramen and passes through the parotid gland, where it divides into five major branches.
Though it passes through the




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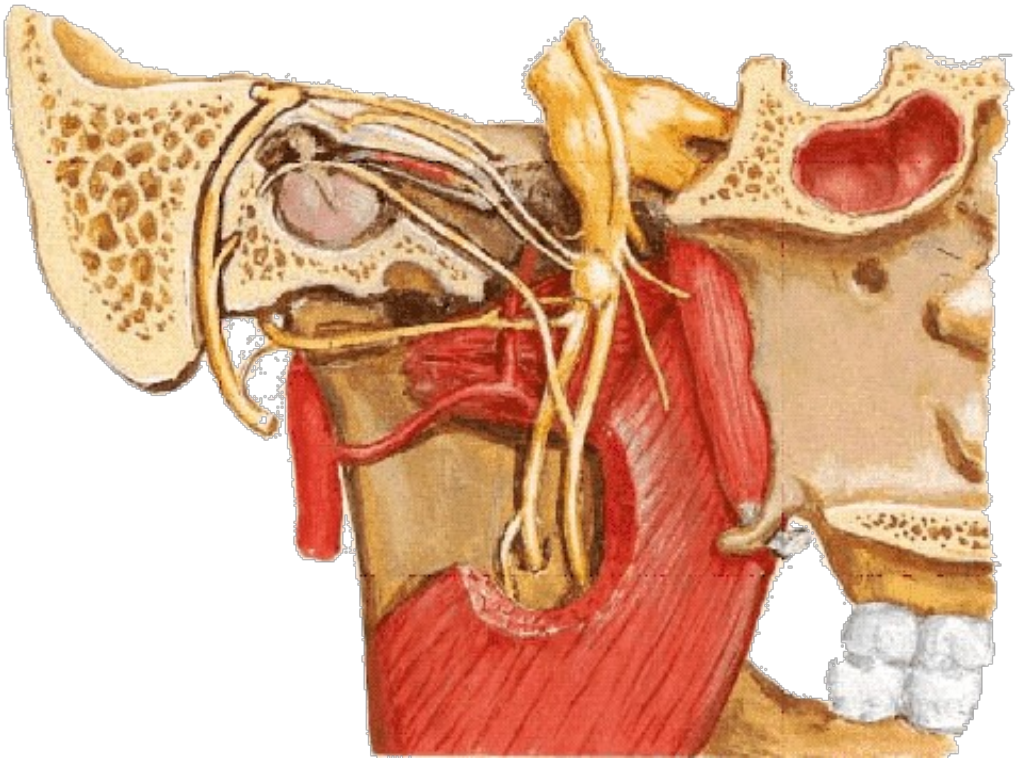

A
Maxillary artery and vein
Transverse facial artery and vein
Superficial temporal artery and vein
Temporal branches
Zygomatic branches
Facial nerve [VII]
Posterior auricular artery
Parotid duct
Retromandibular vein
Buccal branches
Parotid gland
Buccinator
External carotid artery
Masseter
External jugular vein
Marginal mandibular branches
Cervical branches

Course: leaves skull through internal acoustic meatus, facial canal and stylomastoid foramen, it then enters parotid gland where it divides into five branches which supply facial muscles

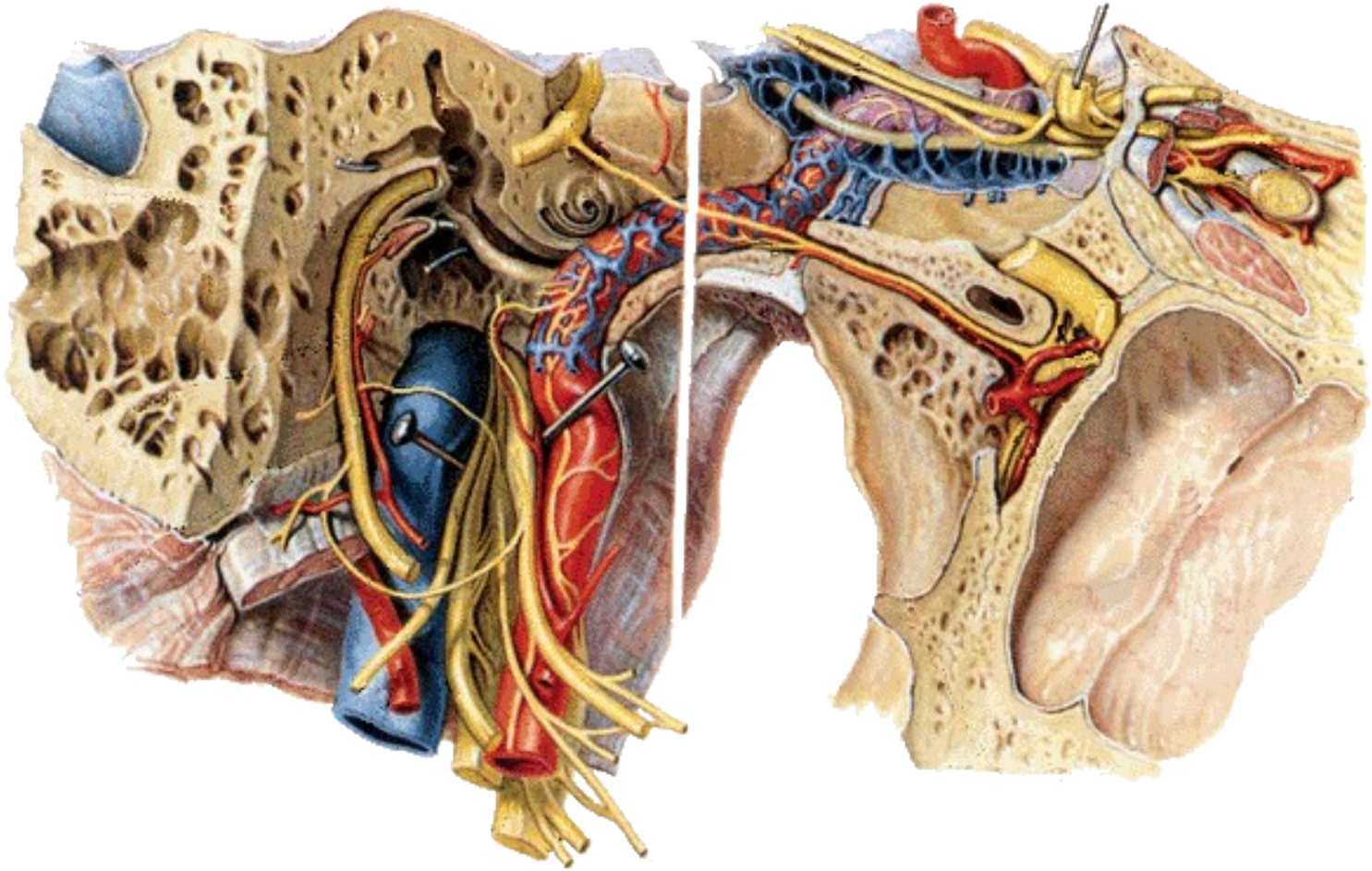


Branches within the facial canal

- **Chorda tympani** : joins lingual branch of mandibular nerve
  - To taste buds on anterior two-thirds of tongue
  - Relayed in submandibular ganglion, the postganglionic fibers supply submandibular and sublingual glands



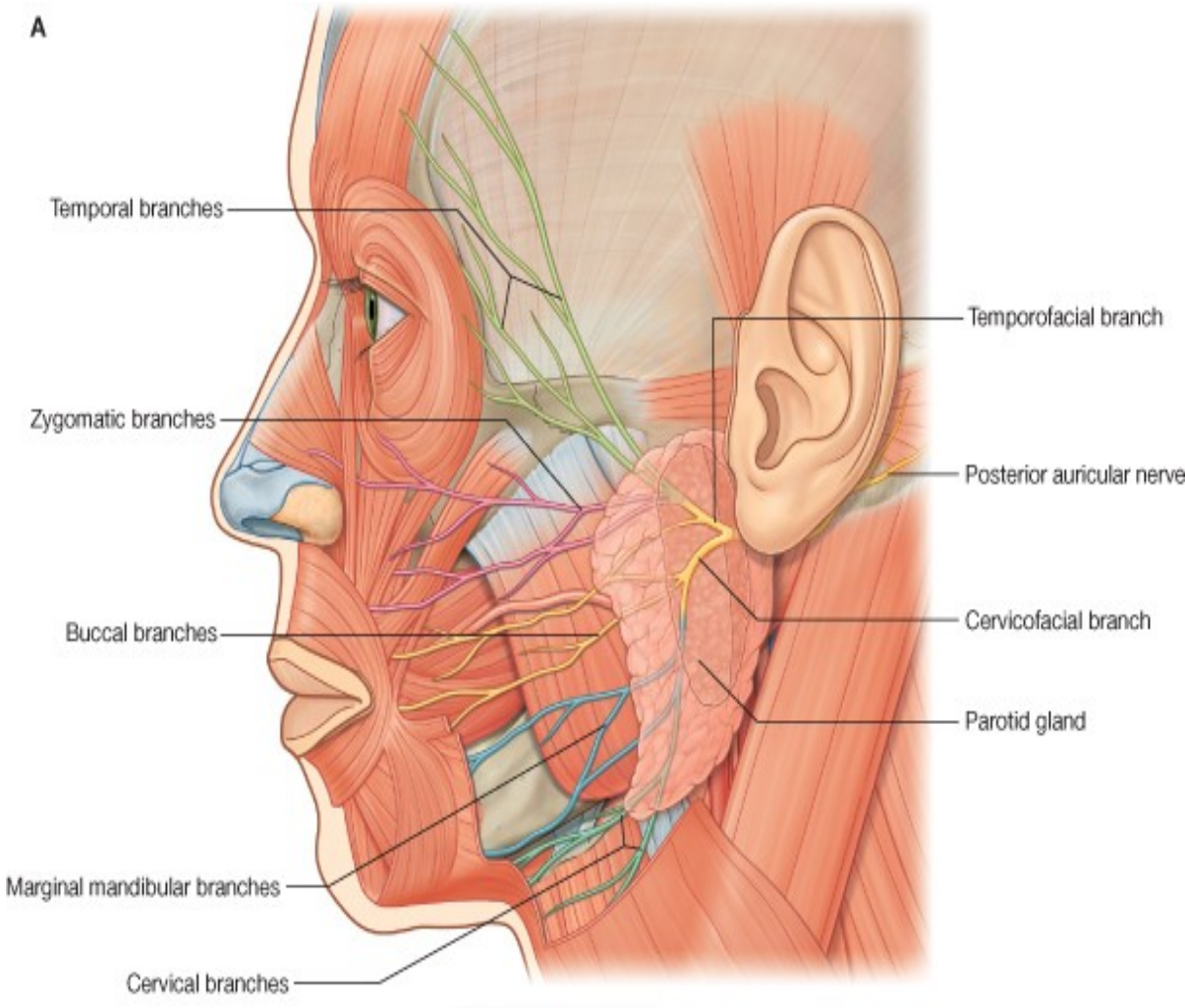
- **Greater petrosal nerve** GVE fibers pass to **pterygopalatine ganglion** and there relayed through the zygomatic and lacrimal nerves to lacrimal gland
- Stapedial nerve : to stapedius



- The branchiomotor component of the facial nerve controls the muscle of facial expression through five branches which are distributed in the superfical fascia of the head and neck. These branches include

Branches outside of facial canal

- Temporal - auricular and fronto-occipitalis muscles
- Zygomatic - muscles of the zygomatic arch and orbit
- Buccal - muscles in the cheek and above the mouth
- Mandibular - muscles in the region of the mandible
- Cervical - the platysma muscle





Injury to the facial nerve